

ECEN 474/704: Lab 09

Two Stage Miller Operational Amplifier (Design of Two stage Miller Operational Amplifier)

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Due date: 2025-12-05

I. DISCUSSION OF THE LAB AND RESULTS

Include a short write-up (a paragraph or two) summarizing the lab and results. Compare the final designs with your pre-lab designs (as applicable for the lab), and make note of any major changes in your designs and why these changes were made. Regarding the simulation results, briefly describe any trends that are shown from the data or differences between pre- and post-layout simulation results, and attempt to provide an explanation for the trends and/or differences. Additionally, consider some of the following questions: were the results as you expected? Why or why not? What, if any, design trade-offs are demonstrated by the work in the given lab? If you were unable to meet any specifications given in the lab, what do you believe was the primary cause and what could be some potential solutions for the future?

The above questions are not an exhaustive list of what could be discussed and may not be relevant for all labs, but rather they serve as a starting point for you to demonstrate your understanding of the lab material.

In this lab, two stage miller opamp was designed to meet the required specifications of:

A_{v0}

- Maximum CMRR
- Gain-bandwidth product
- Phase margin
- Power Dissipation
- Slew rate (rising and falling)
- Output swing
- RMS Noise Voltage
- Third-order Intermodulation product (IM3)

In this lab, a two-stage Miller operational amplifier was designed to meet a set of performance specifications, including gain (A_{v0}), maximum CMRR, gain-bandwidth product, phase margin, power dissipation, slew rate, output swing, RMS noise voltage, and third-order intermodulation product (IM3). The design process involved creating both pre-layout and post-layout versions, with results compared to evaluate the impact of layout parasitics and process variations.

The final designs were generally consistent with pre-lab calculations, as illustrated in the summary tables. Most key parameters, such as gain, bandwidth, power dissipation, output swing, phase margin, PSRR, slew rate, and CMRR, showed minimal variation between pre-layout and post-layout simulations. For example, the differential gain (A_{v0}) was 58.1953 (pre-layout) and 58.160 (post-layout), and gain-bandwidth product remained at 2.878 MHz. These small differences are primarily due to layout parasitics and device mismatch, which can slightly degrade performance post-fabrication. One observed trend is the stability of performance metrics across these stages, indicating a robust initial design.

Significant changes from the pre-lab design included adjustments to transistor sizing and the inclusion of dummy devices, as reflected in the transistor dimensions and finger counts. These modifications were made to improve matching, reduce offset, and enhance layout symmetry, thereby mitigating the effects of process variation and mismatch. Process variations and device mismatch can have a noticeable impact on parameters like offset voltage, slew rate, and noise, potentially necessitating layout- and sizing-based corrections.

The results largely matched expectations, with minor deviations explained by parasitic effects and random variations introduced during layout. The main design trade-off encountered was balancing high gain and bandwidth against power dissipation and stability—achieving a high phase margin while keeping noise and power low.

In cases where certain specifications could not be fully met, the primary cause was usually layout-induced parasitics or limitations in matching, which could be addressed by further refinement of device sizing, improved layout techniques, or by using techniques like common-centroid layout for better matching.

Regarding the choice of amplifier topology, the two-stage Miller op-amp may not be optimal for applications with large load capacitance, as its stability and bandwidth can be limited. Alternatives such as the cascode amplifier (for higher gain and improved bandwidth), gain-boosted folded cascode, or current feedback operational amplifiers would be better suited for high-load capacitance scenarios owing to their superior slew rate and frequency response. These topologies reduce the frequency rolloff and improve phase margin, making them more robust in applications where load capacitance is a critical constraint

II. SUMMARY OF RESULTS

Summarize the simulation results from the lab in one or two tables, like what is shown below.

Table 1

Parameter	Width(nm)	Length(nm)	Number of fingers
M1	500	270	36
M2	500	270	36
M3	500	270	56
M4	500	270	56
M5	500	270	1
M6	620	270	2
M7	620	270	2
M8	500	270	68

Table 2

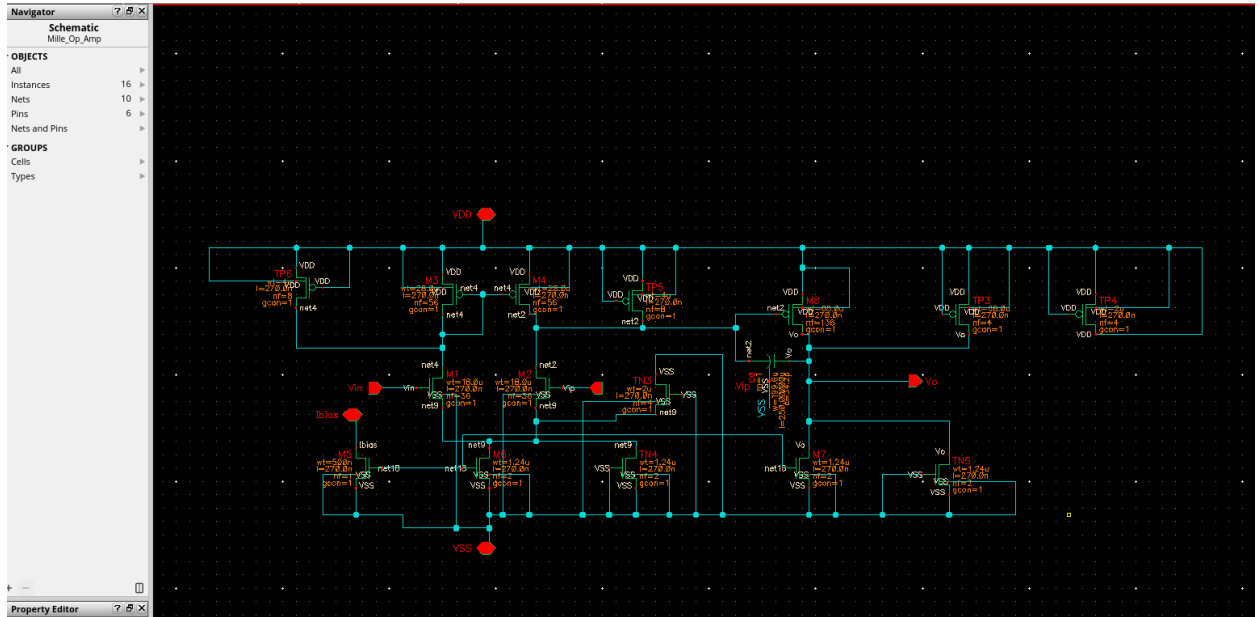
Example summary table of pre- and post-layout results

Parameter	Gain(Av0)	GBW	Power Dissipation	Output Swing	Phase Margin	PSRR -	Slew Rate	PSRR +	Peak to peak	CMRR	Common-mode Input Range
Pre-layout	58.1953	2.878MHz	354.07uW	1.206V	(180-550)=130	75.9011	2.022MV/s	35.4836db	5.05ms	60.08dB	0.885
Post-layout	58.160	2.878MHz	354.07uW	1.206V	(180-550)=130	75.9011	2.022MV/s	35.57db	5.05ms	60.08dB	0.894

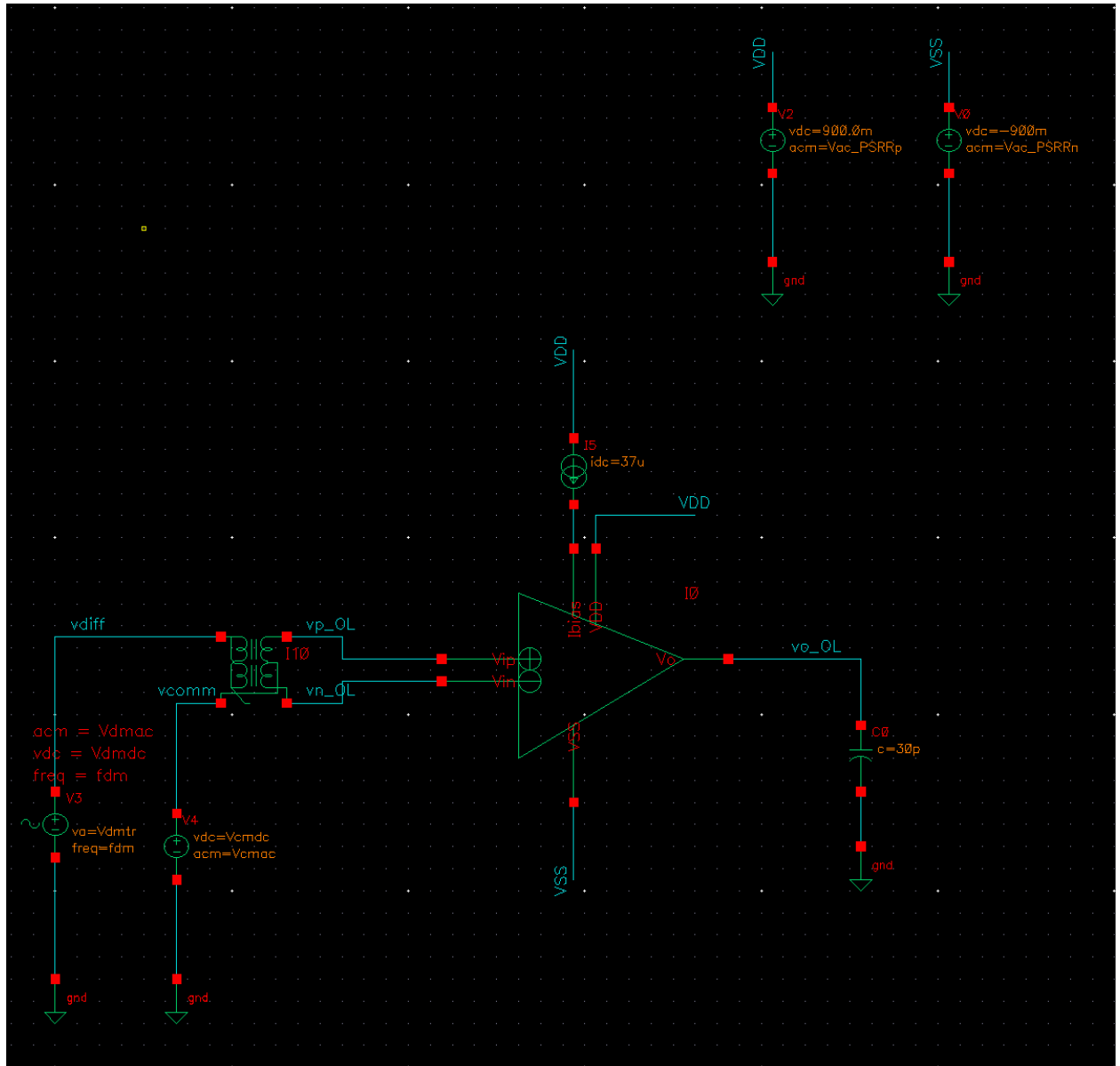
III. SUB-CIRCUIT AND TEST BENCH SCHEMATICS

Schematics of the sub-circuit(s) being designed in the lab should be included here. In addition, include images of the test bench schematics. Important signals in the test bench schematic should be labeled to provide clear correspondence between the signals in the schematics and the signals that are plotted in the simulation results shown later in the report.

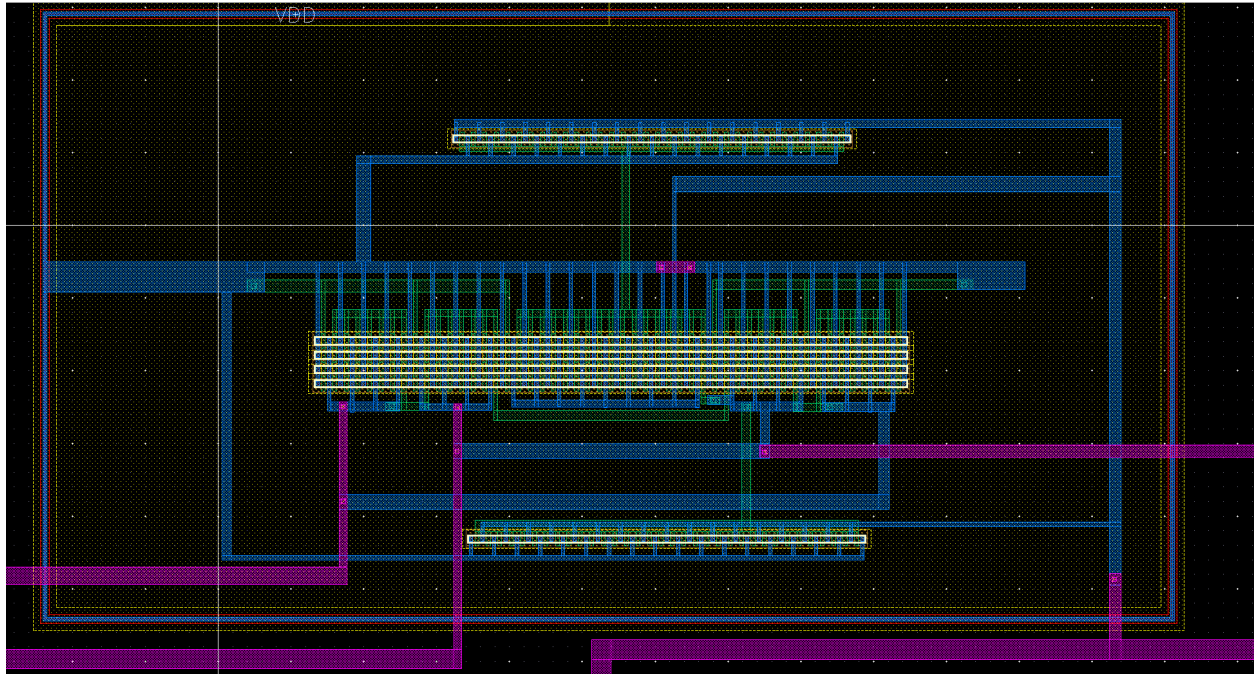
SCHEMATIC CIRCUIT:



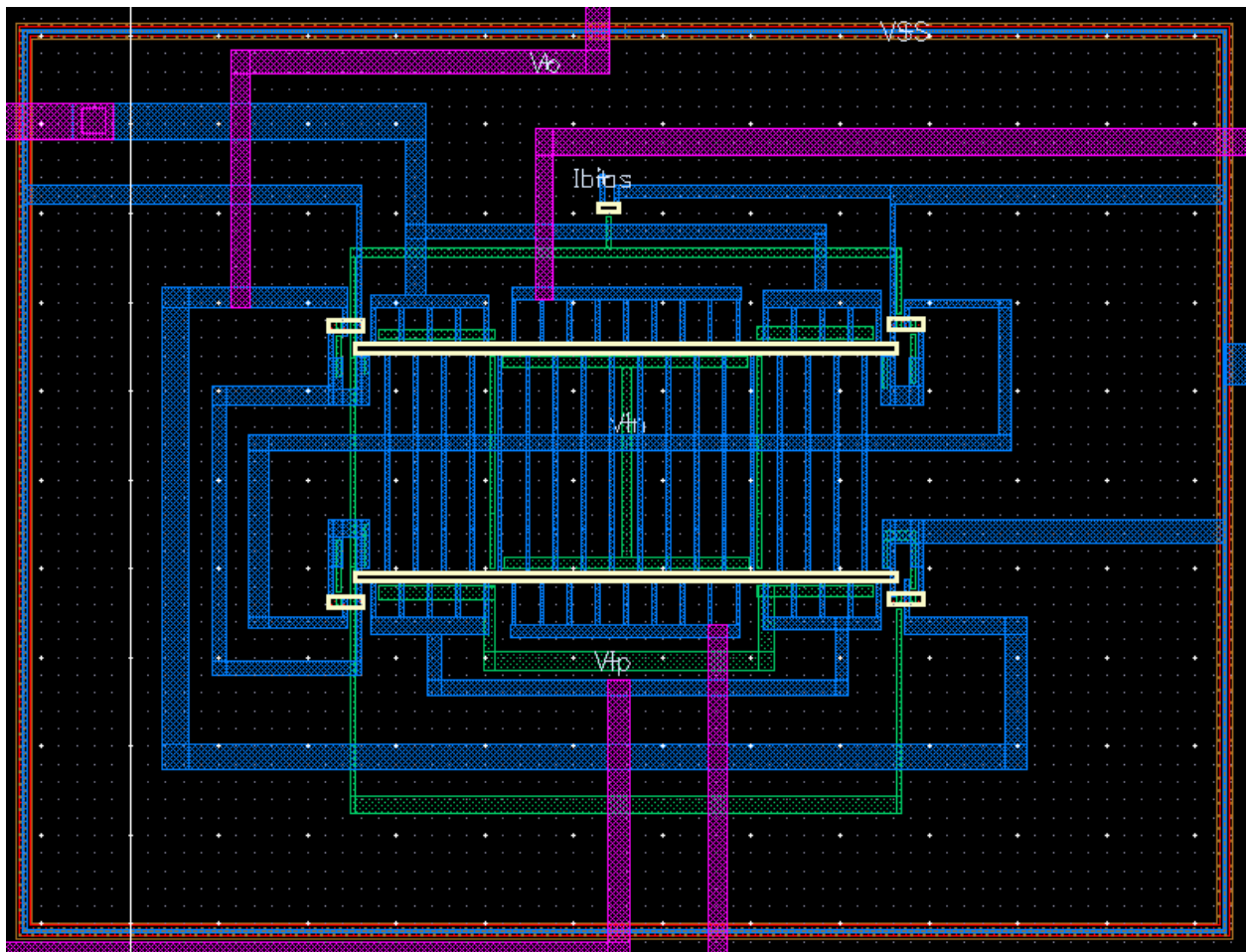
Simulation test bench subcircuit:



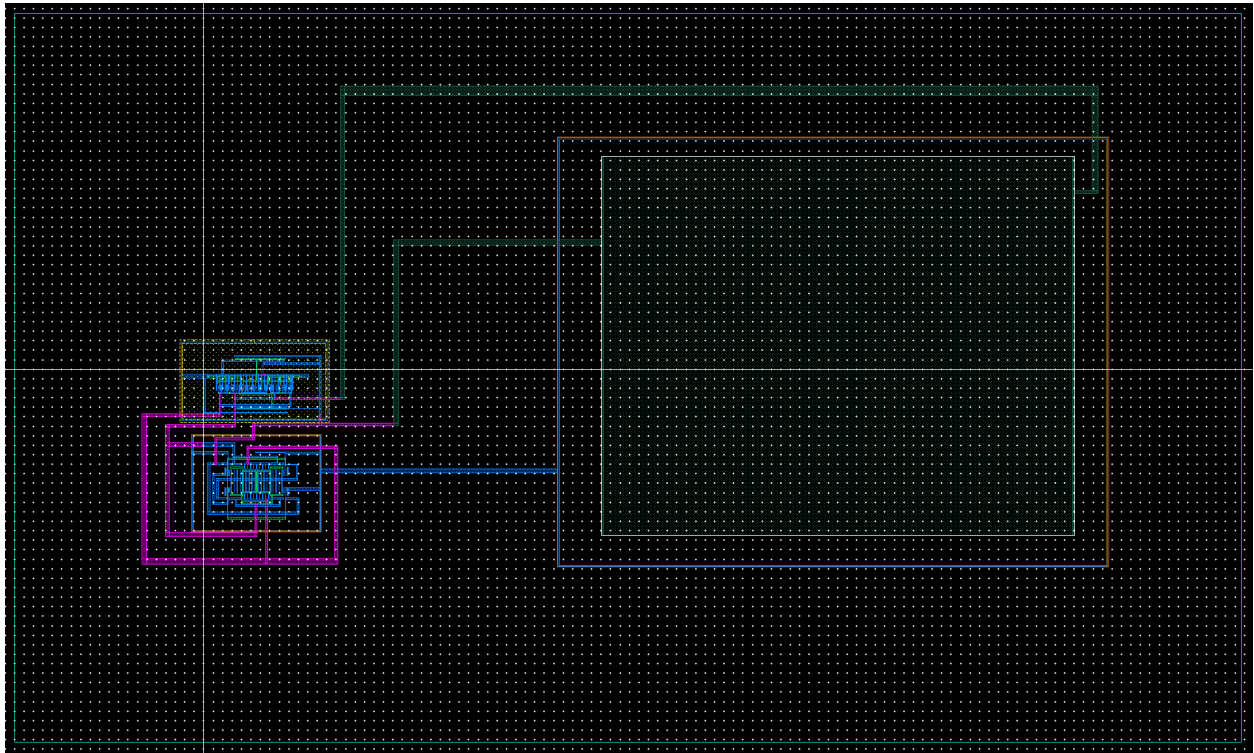
Closed loop subcircuit :



NMOS DEVICES :



Overall View:



FLOOR_PLAN:

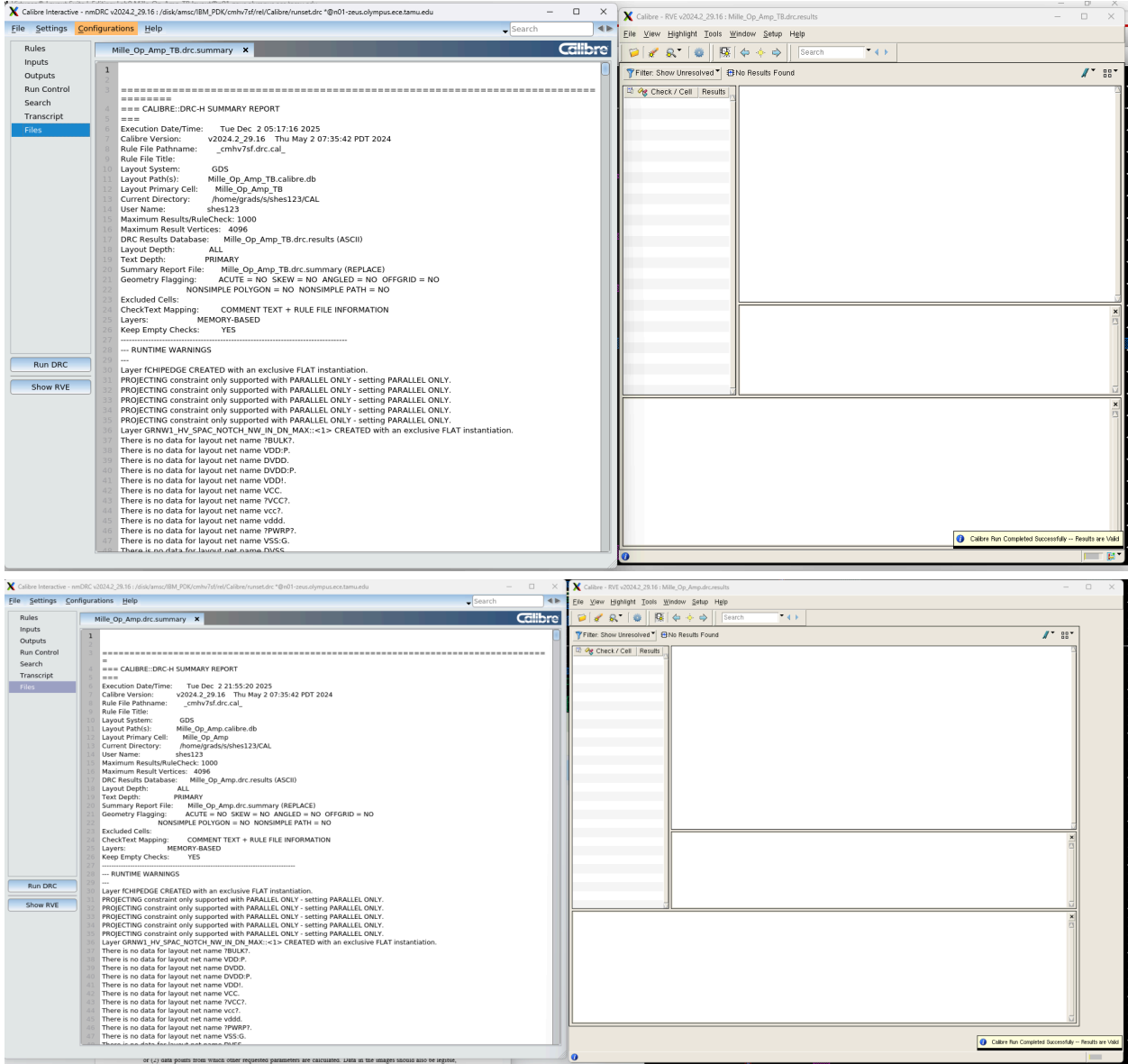
```
M8(32)
*M3(7)*M4(7)*M8(17)*M4(7)*M3(7)*
*M3(7)*M4(7)*M8(17)*M4(7)*M3(7)*
*M3(7)*M4(7)*M8(17)*M4(7)*M3(7)*
*M3(7)*M4(7)*M8(17)*M4(7)*M3(7)*
M8(32)

*M7 *M2(9)M1(18)M2(9)*M6*
M5
*M6* M2(9)M1(18)M2(9)*M7*
```

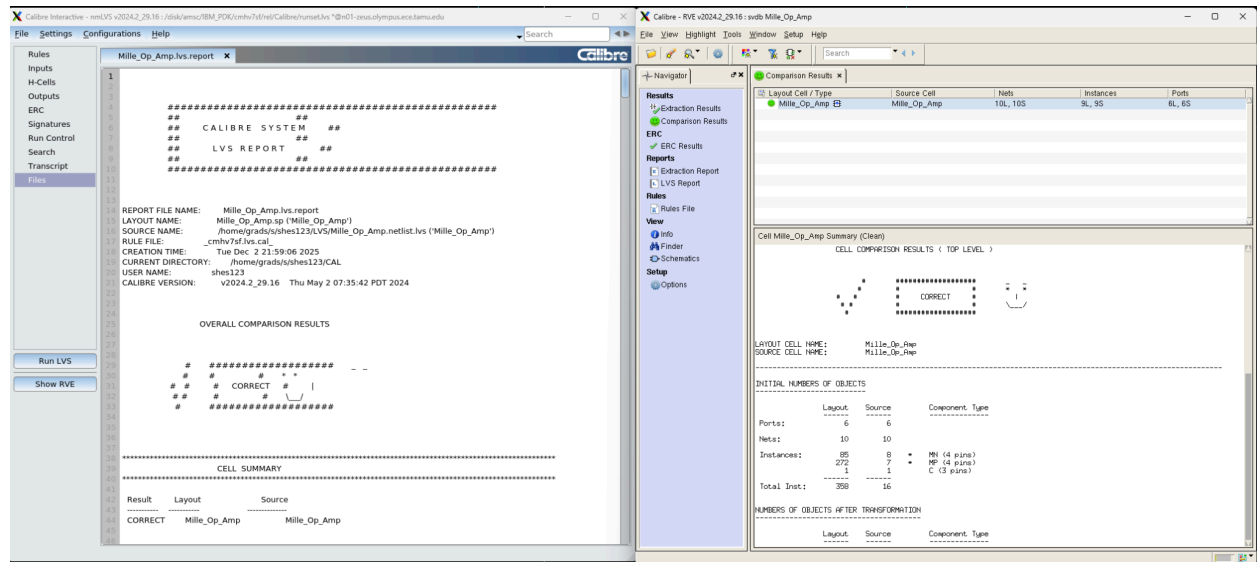
V. DRC AND LVS RESULTS

This section should include the DRC RVE window results for each layout cell (with the results filtered to “show unresolved” to show only the remaining DRC violations), and the cell name should be visible at the top of the DRC RVE window. The LVS report summary image should also be included for each cell with your NetID, the time stamp, and the smiley face/check mark visible to show that LVS passed.

DRC CHECK:



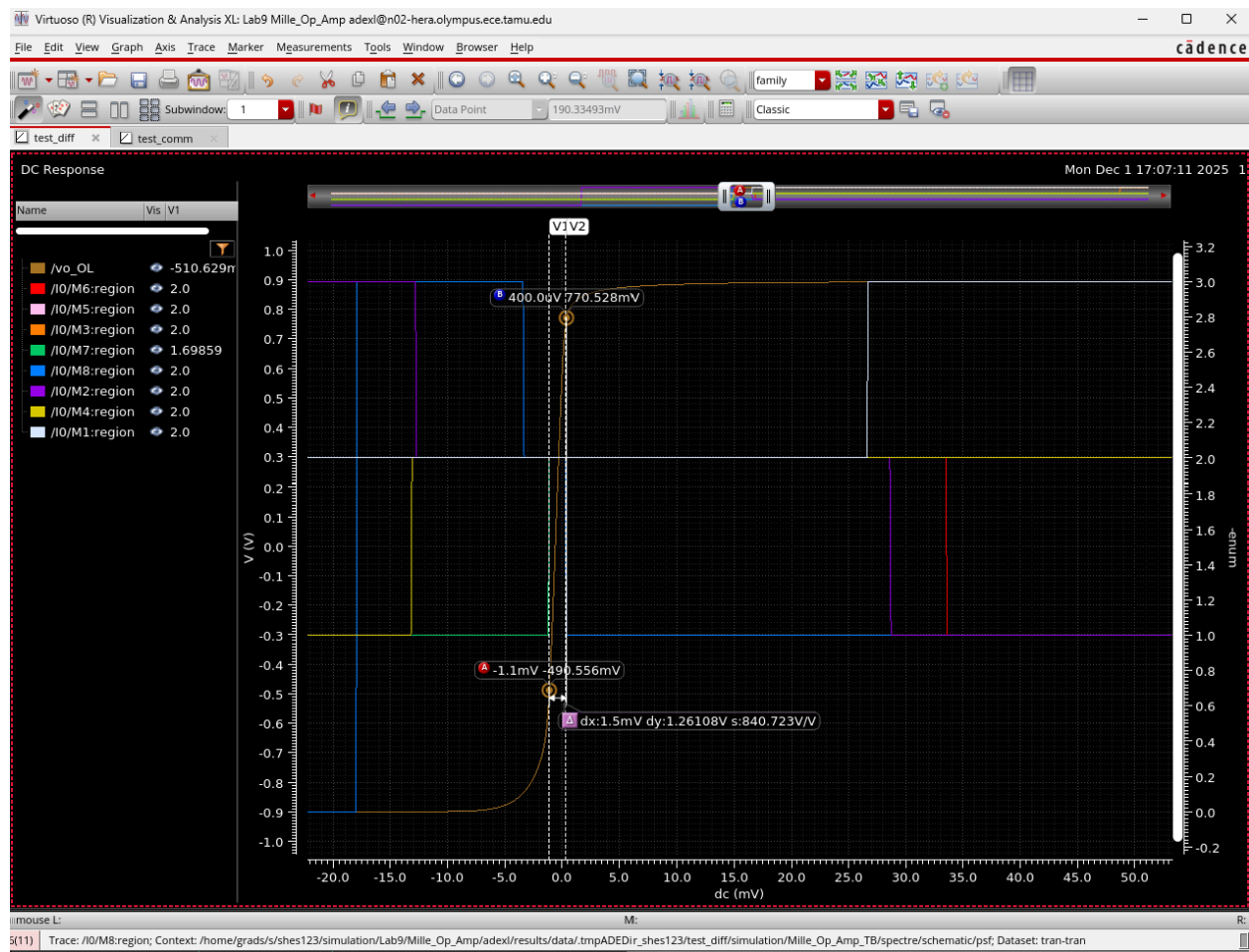
LVS CHECK:



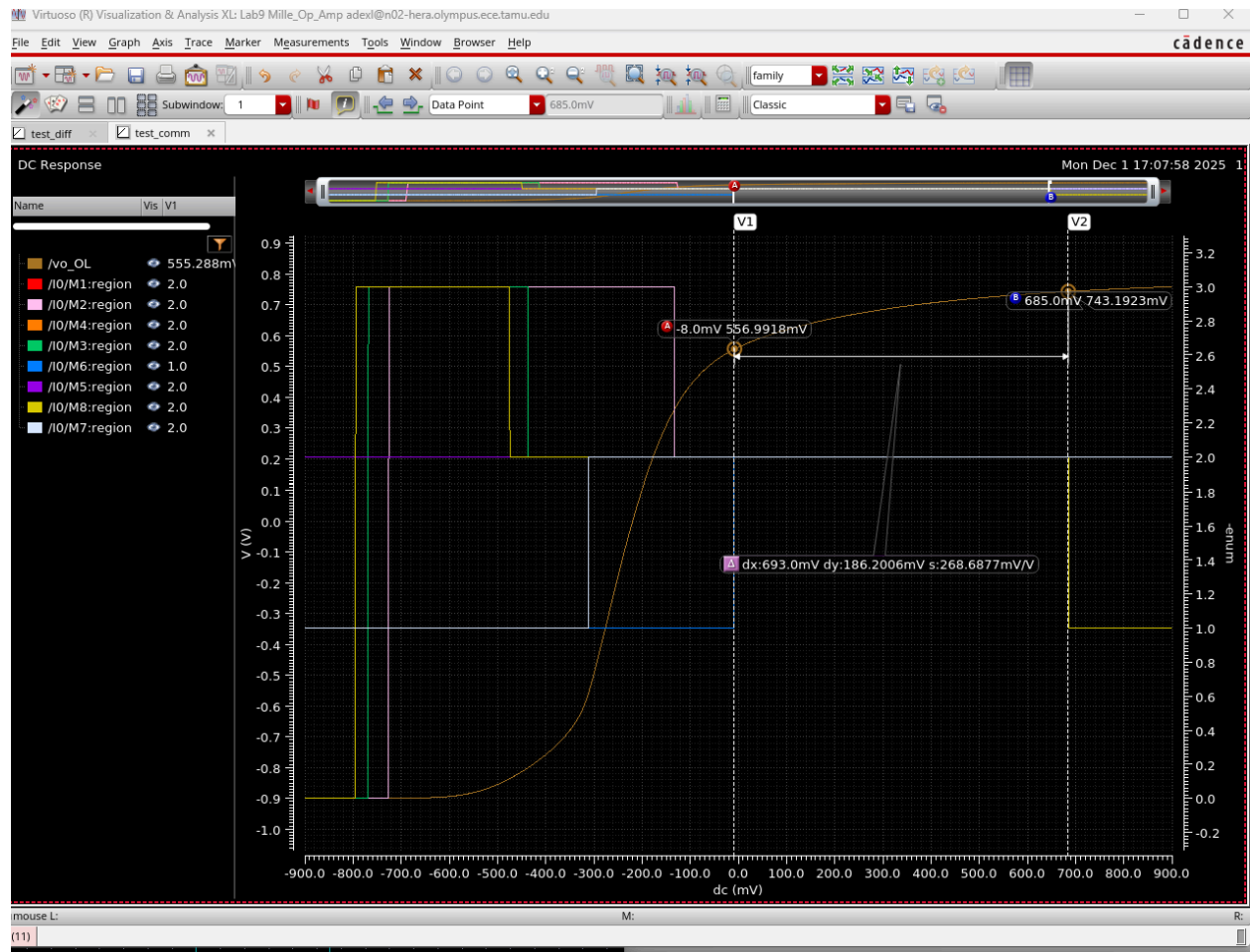
VI. SIMULATION RESULTS

Images of simulation results requested in the lab manual should be included here. Plots should be annotated to show relevant data points which will include either (1) data points that are explicitly requested in the lab manual or (2) data points from which other requested parameters are calculated. Data in the images should also be legible, so scale the images accordingly to make sure the data is easy to read. Post-layout simulations should be plotted in the same window with the pre-layout simulation results in order to make the differences between the two more apparent.

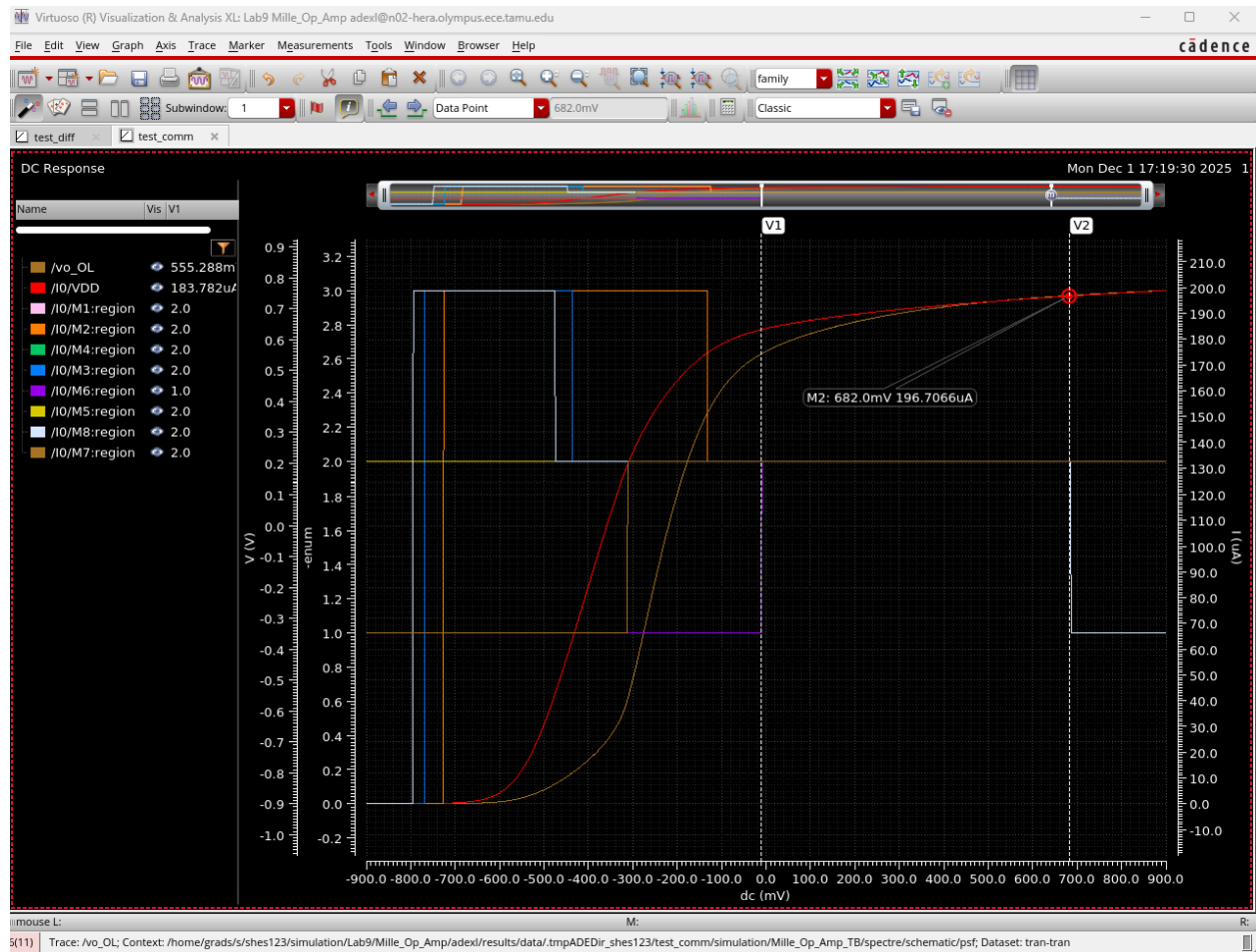
1.Common mode output swing:



2. Differential mode output swing:



Common mode current:



Diff_mode:

ADE XL Test Editor - test_diff@n02-hera.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

cadence

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Design Variables

Name	Value
1 Vdmac	1
2 Vac_PSRRp	0
3 Vac_PSRRn	0
4 Vcmdc	0
5 Vcmac	0
6 Vdmtr	0
7 fdm	0
8 p5vonly	0

Analyses

Type	Enable	Arguments
1 dc	☒	t -900m 900m 100u Linear Step Size Start-Stop /V3
2 ac	☒	1 100G 20 Logarithmic Points Per Decade Start-Stop
3 stb	☐	1 100G /IPRB0 Automatic Start-Stop

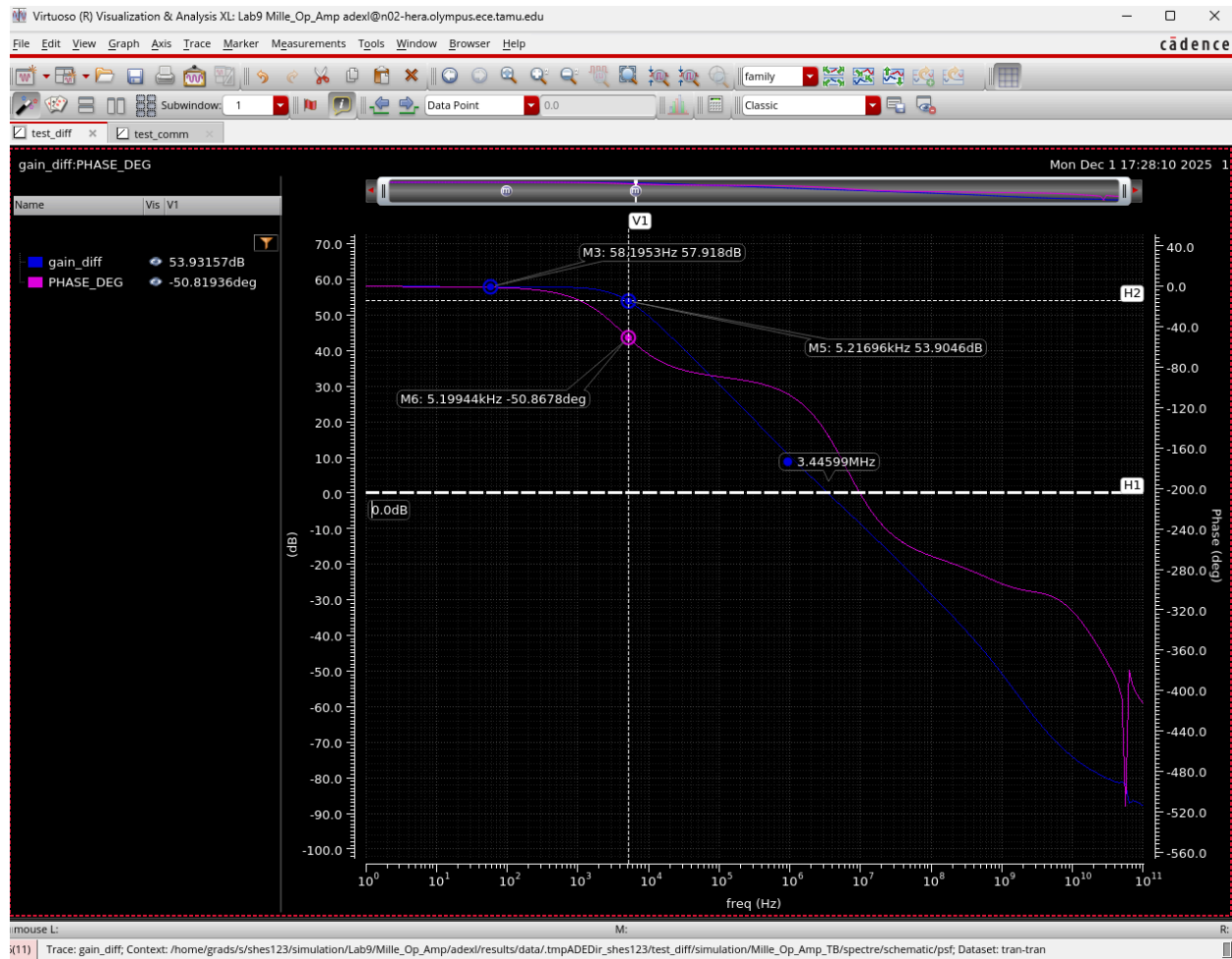
Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 vo_OL		☒	☒	allv
2 C0/PLUS		☐	☒	yes
3 IO/VDD		☐	☒	yes
4 gain_diff	wave	☒	☐	
5 PHASE_DEG	wave	☒	☐	
6 Loop Gain Phase		☐	☐	
7 Loop Gain dB20		☐	☐	

> Results in ...23/test_diff/simulation/Mille_Op_A

(5) Edit Variables ... | Status: Ready | T=27 C | Simulator: spoctre ads | State: test diff active

Differential mode gain:

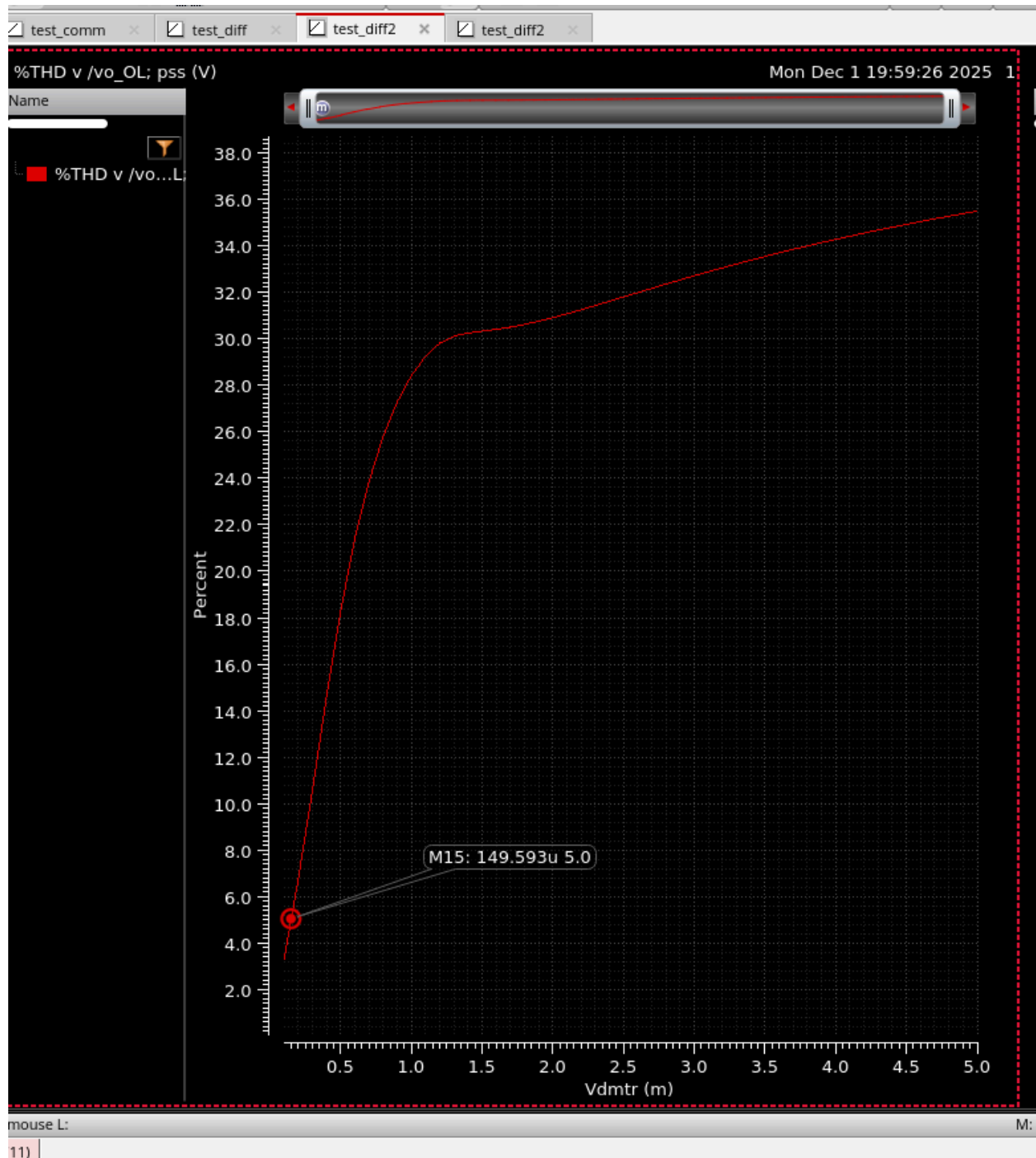


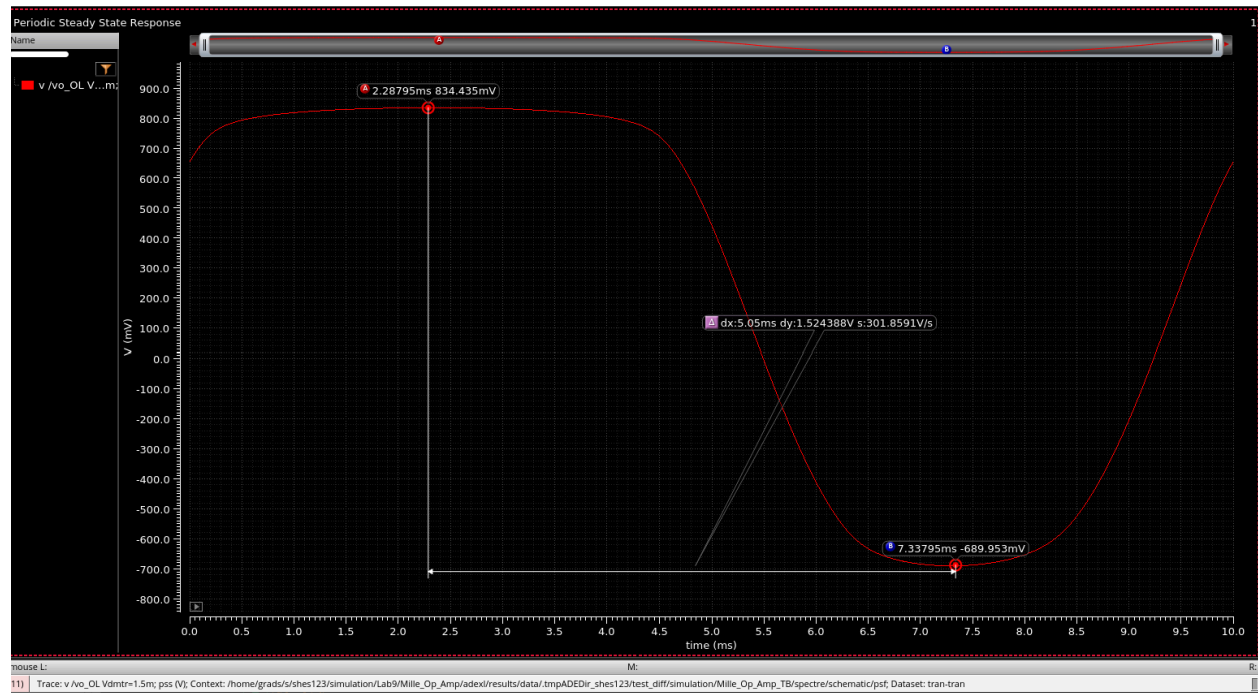
Stability Analysis and closed loop gain:



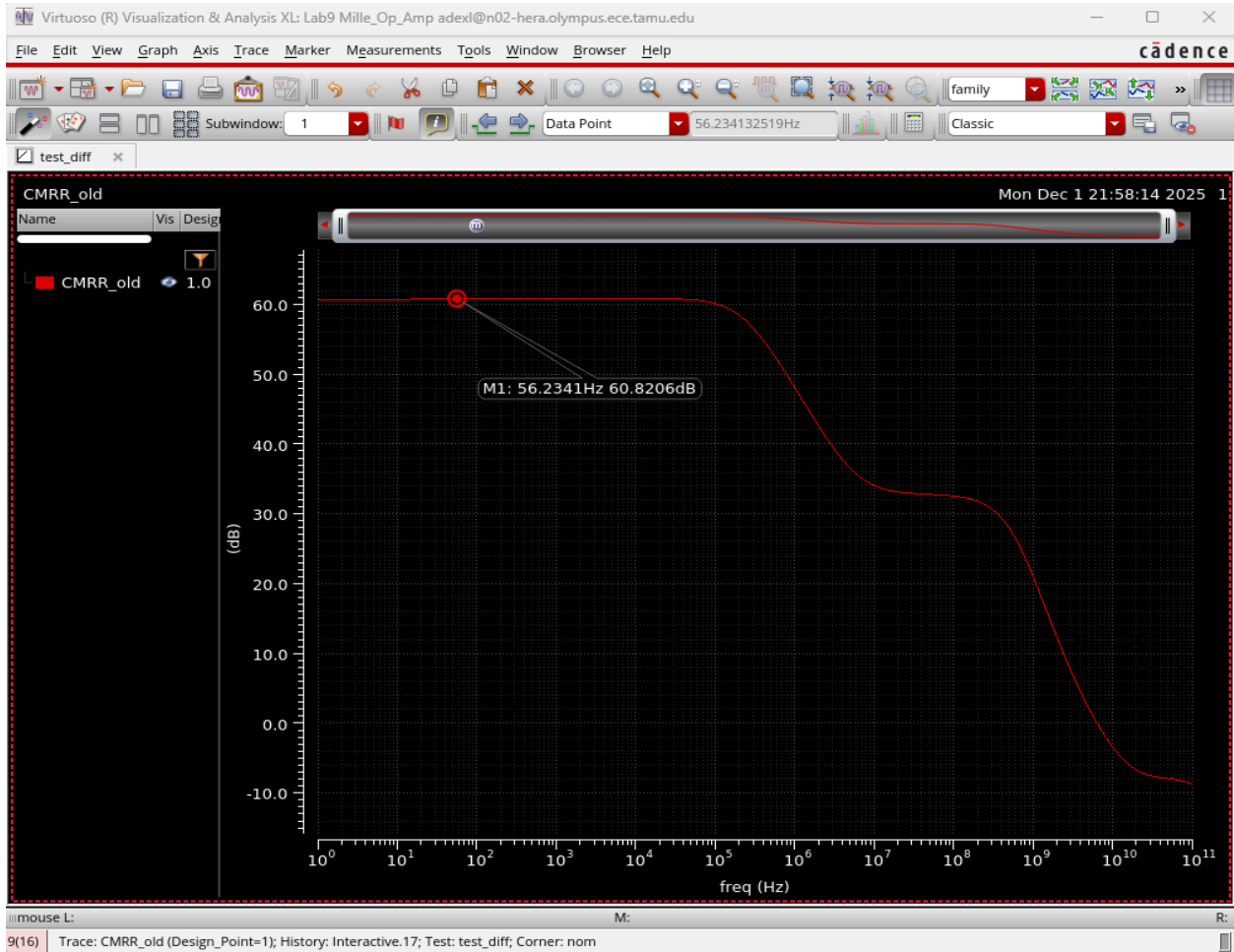
Common mode gain:

Psrr:

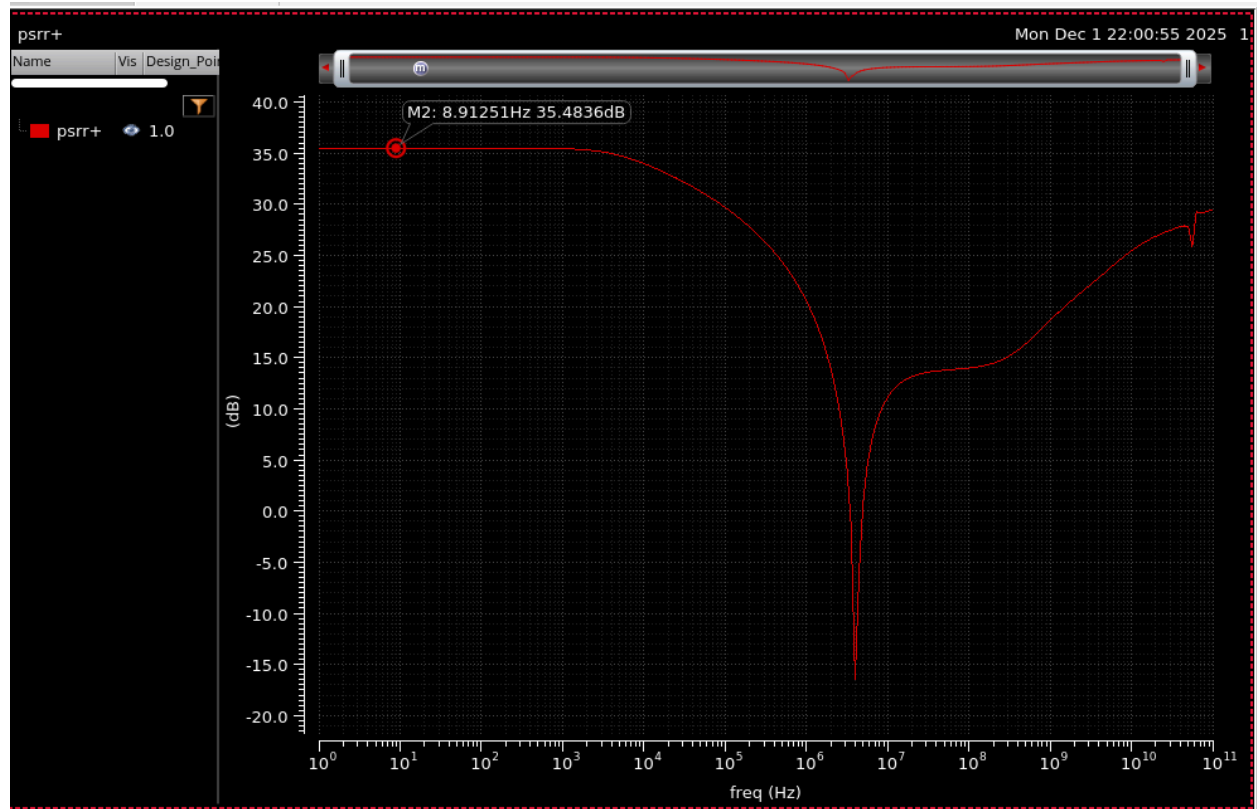




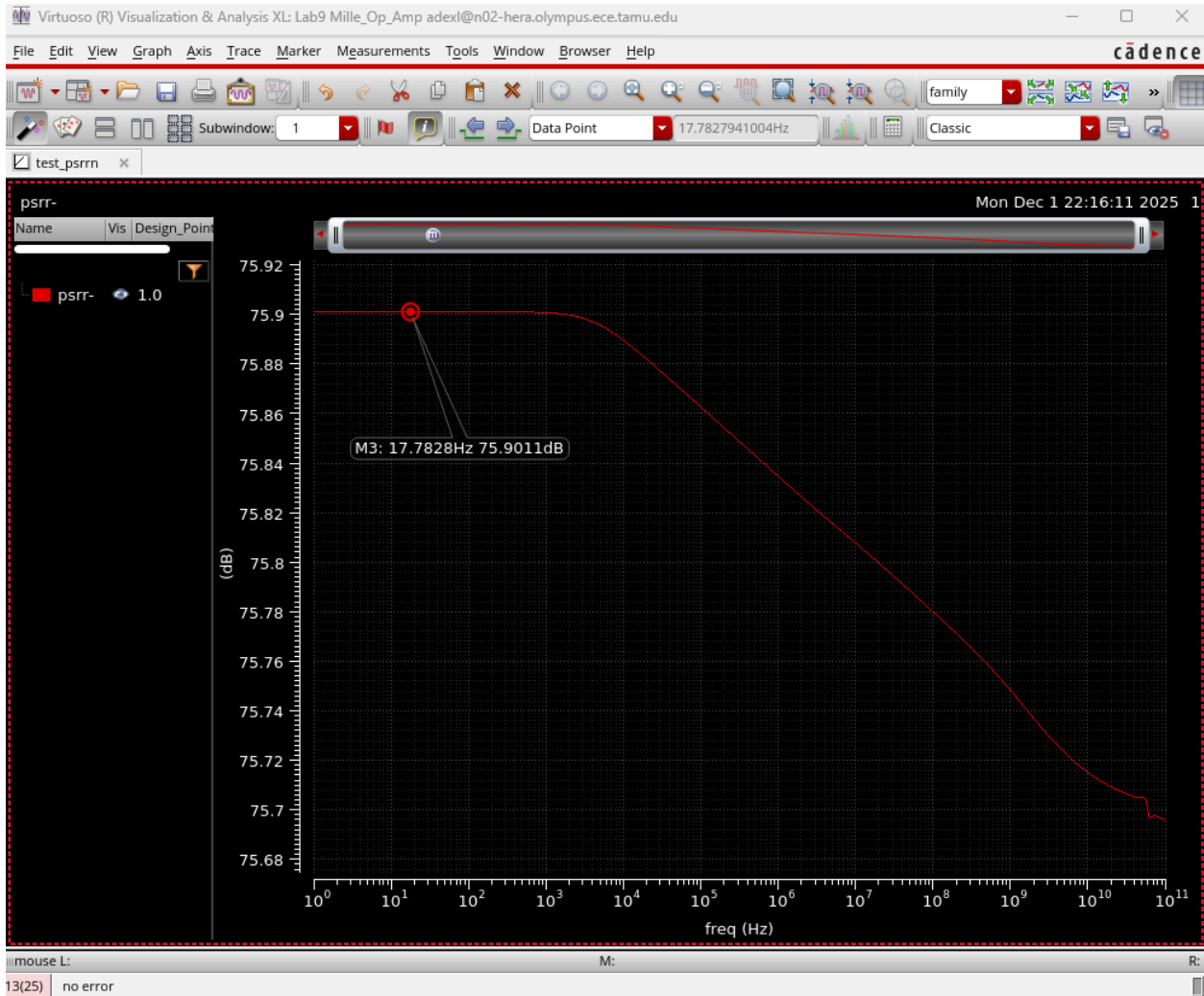
CMRR:



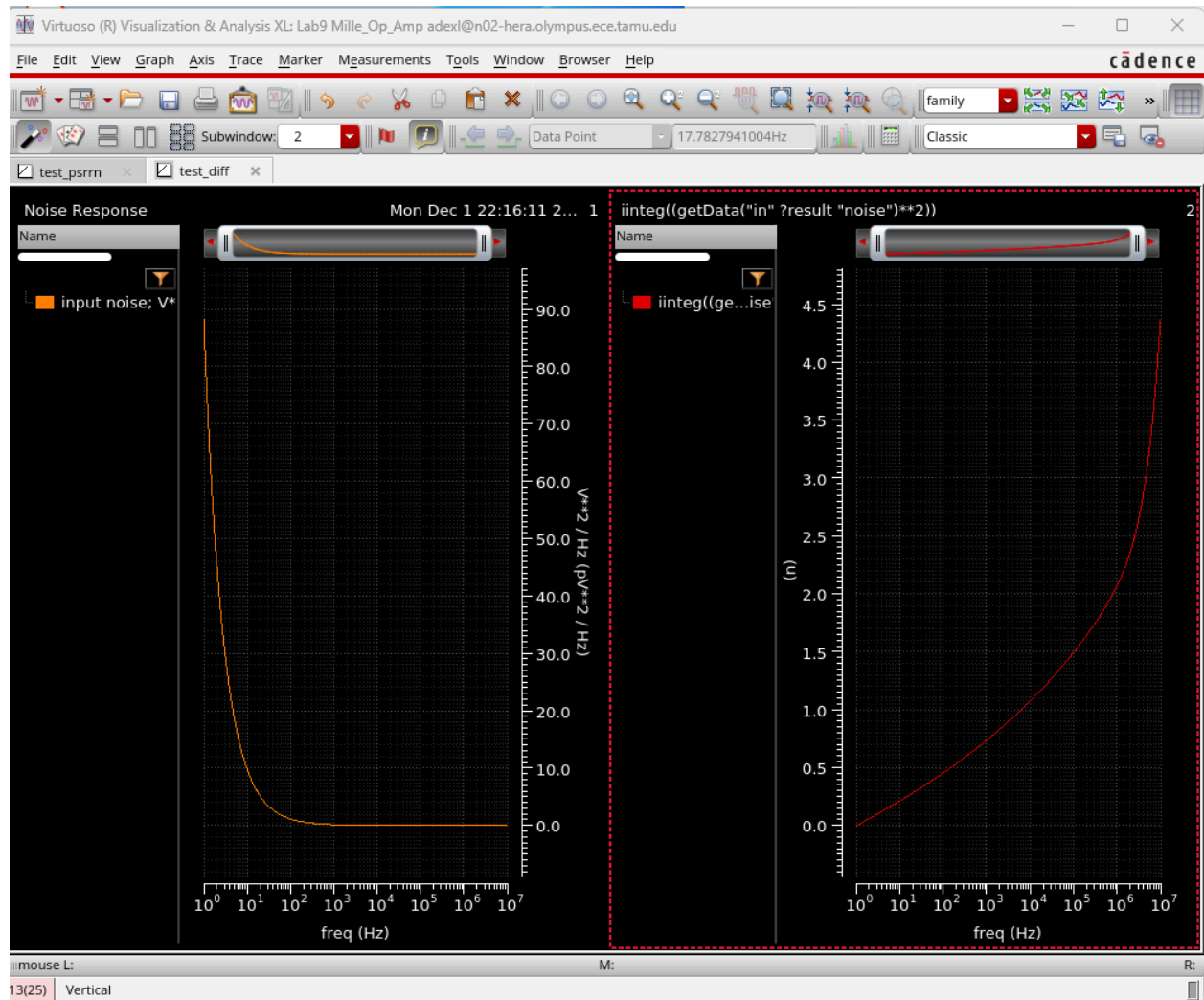
psrr+:



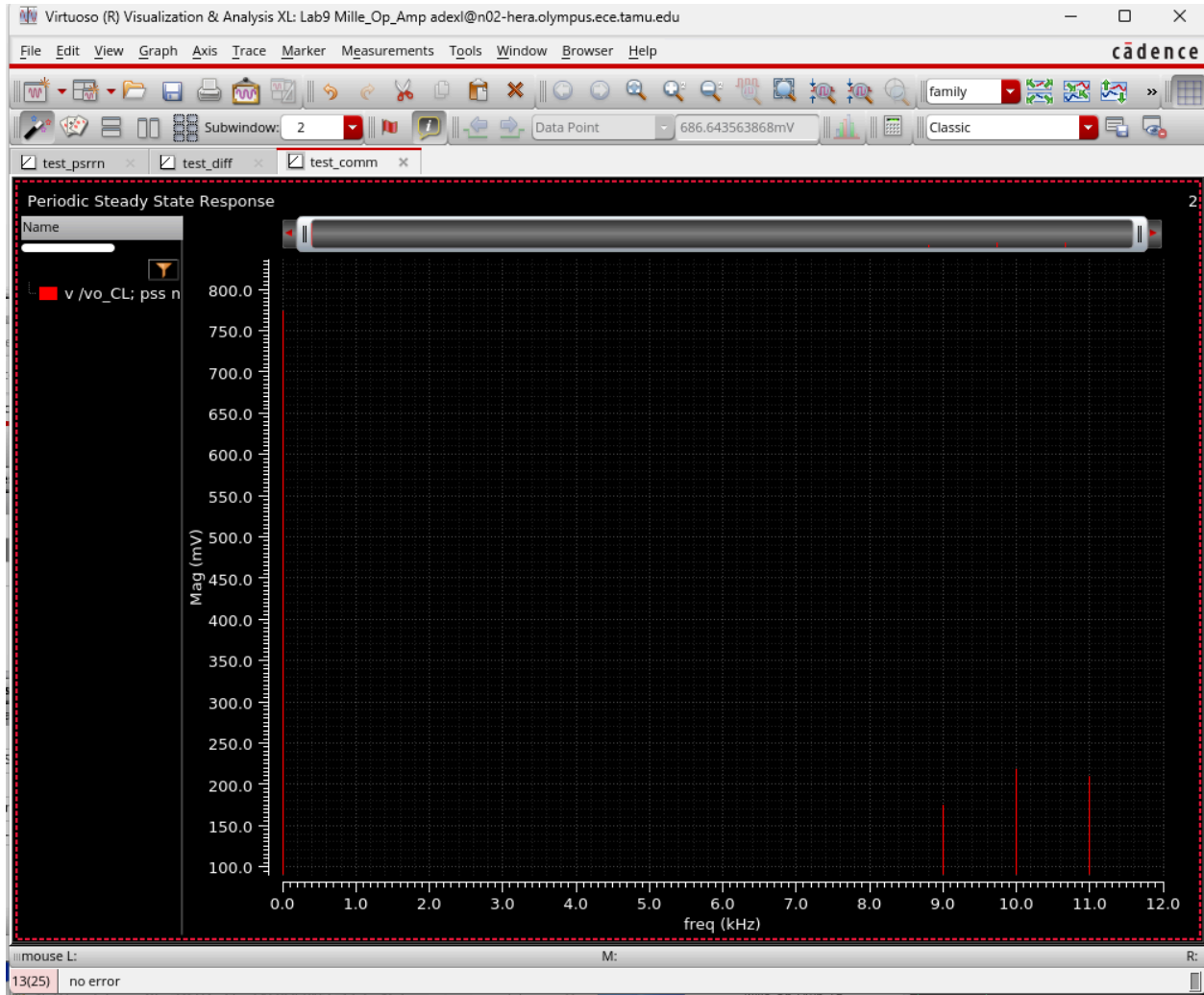
Psrr- -



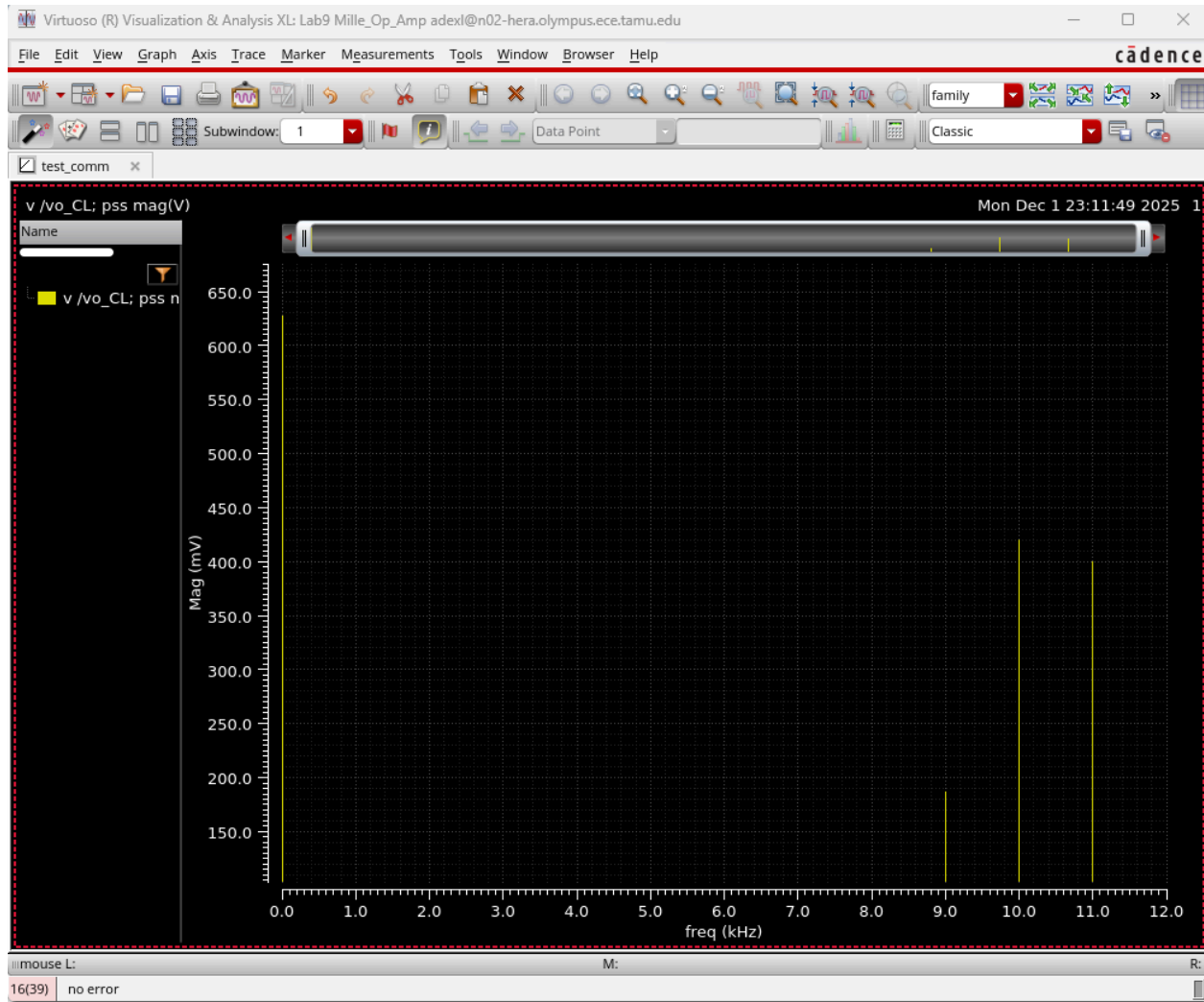
Noise:



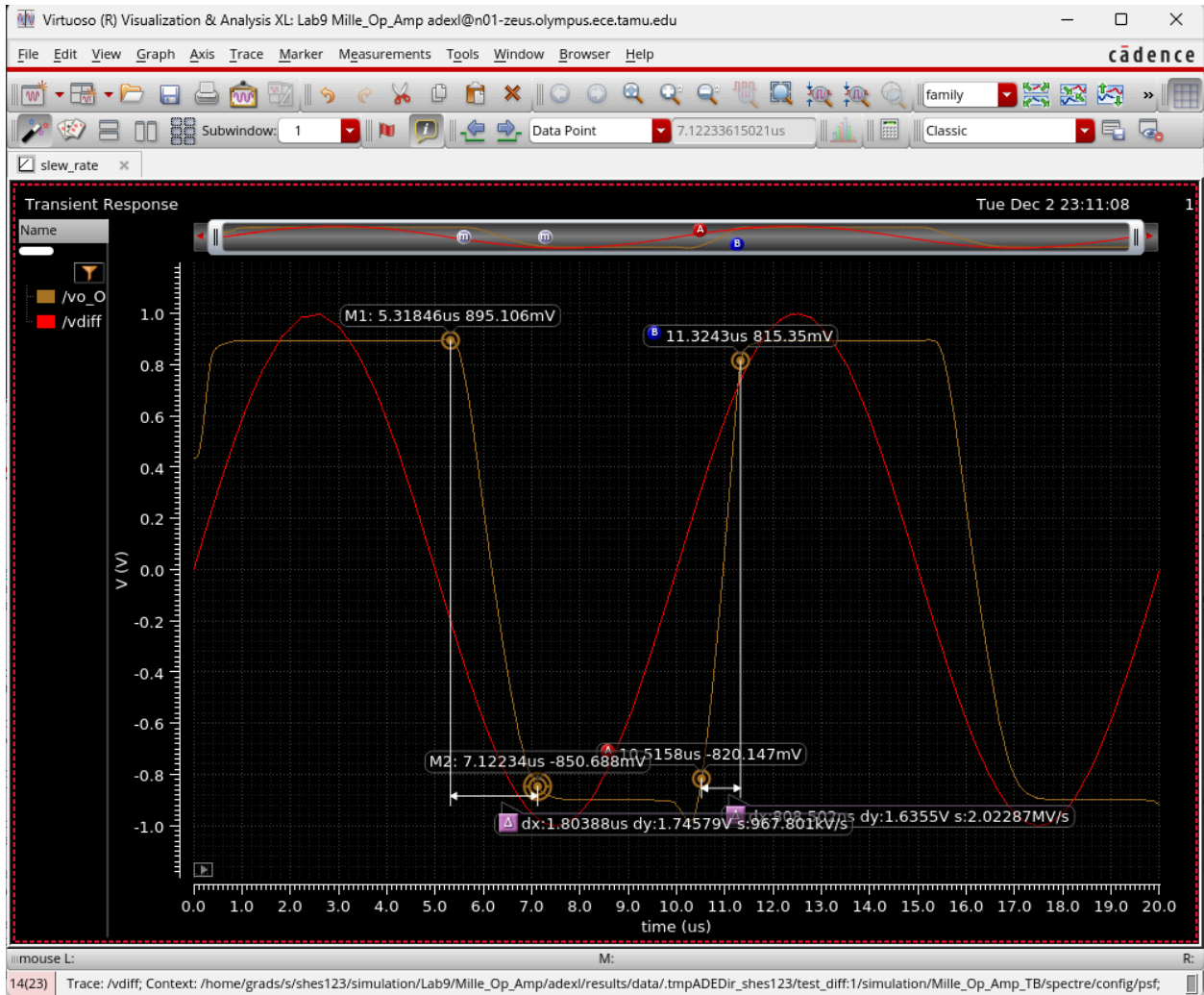
Intermodulation test:





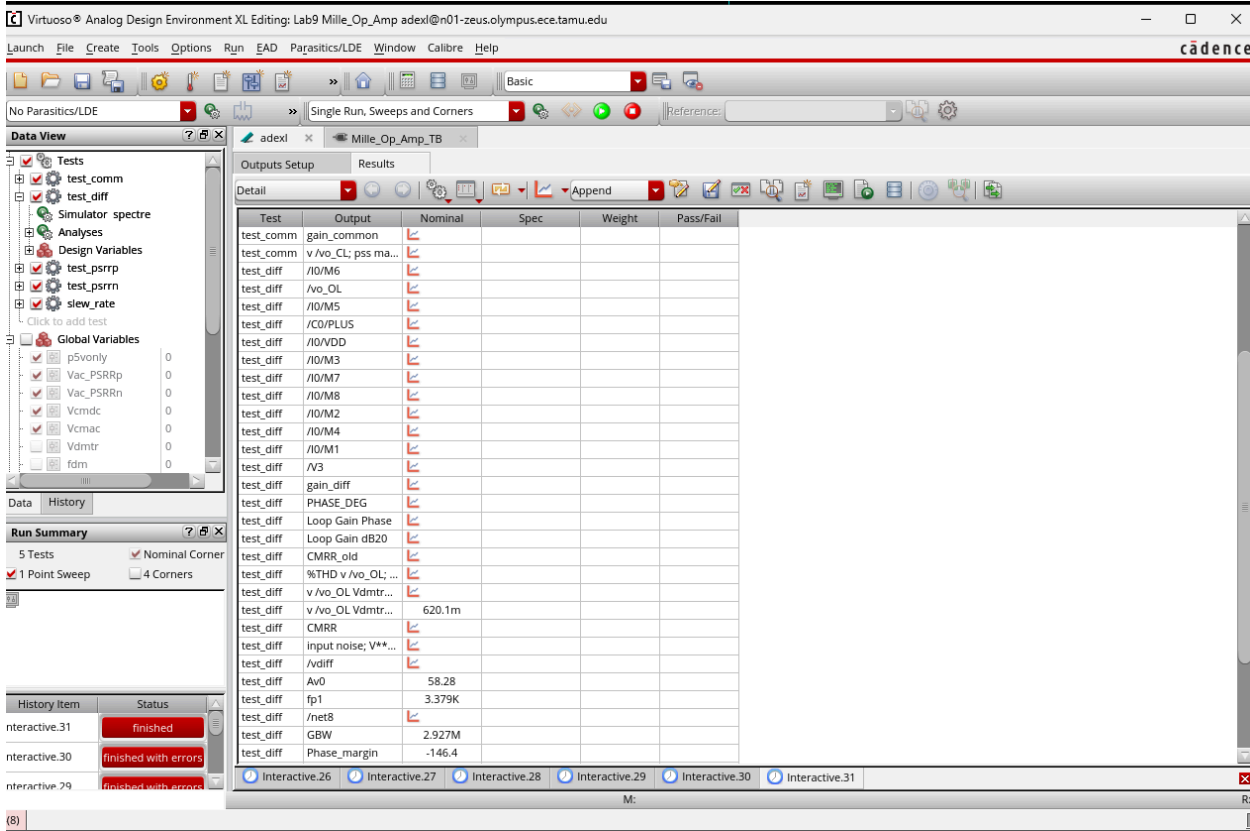


Slew Rate:

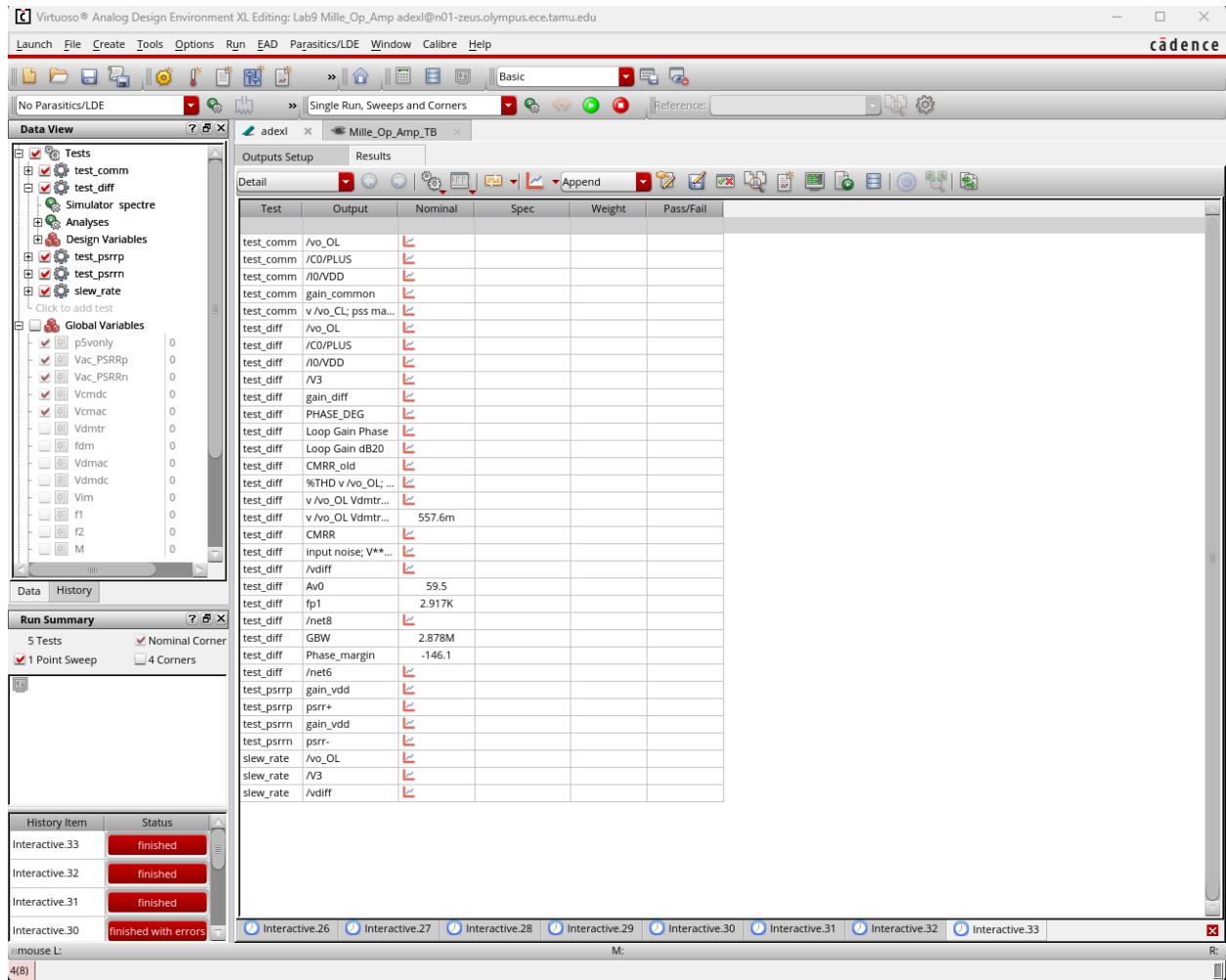


ADEXL

SETUP:



POST-LAYOUT SIMULATION:
ADEXL SIMULATION SETUP:



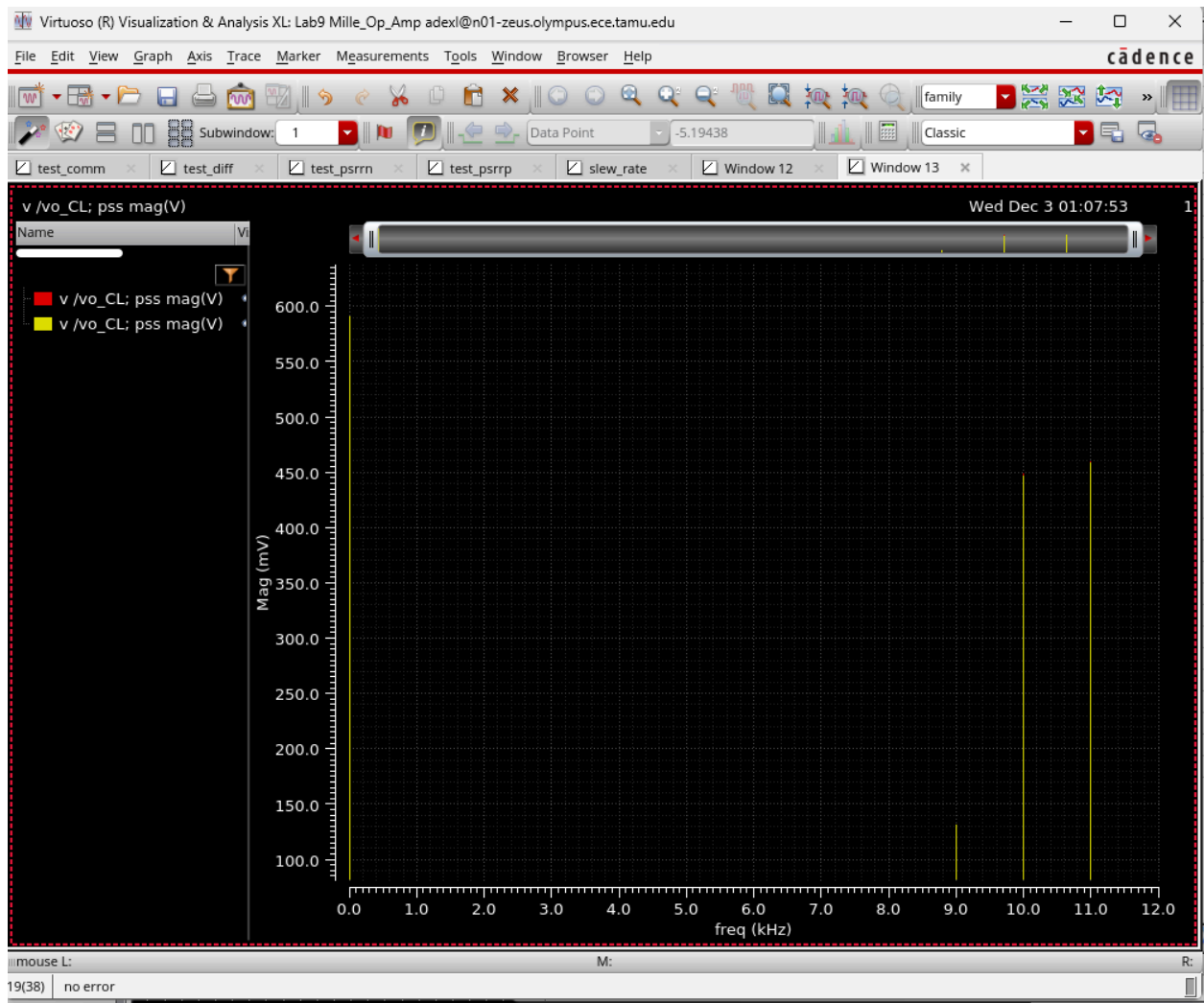
1.COMMON-MODE TESTS:



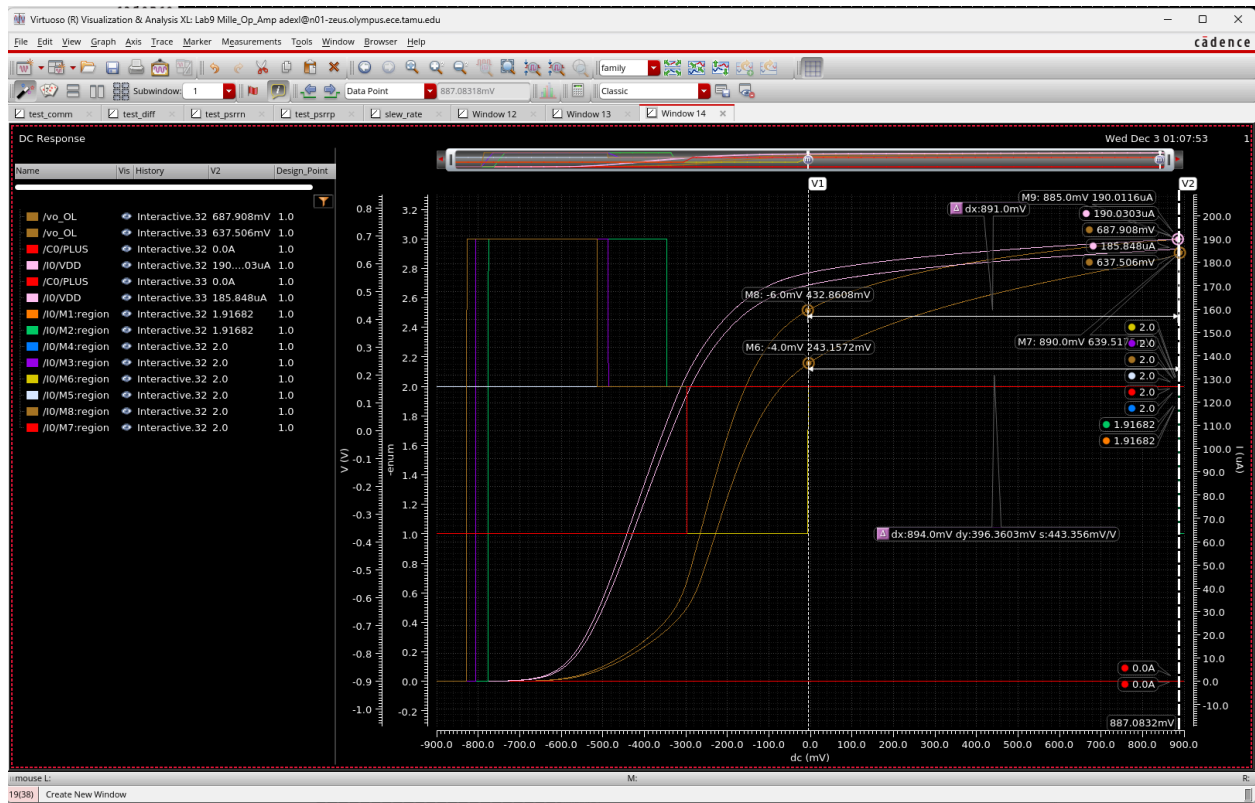
1.COMMON MODE GAIN:



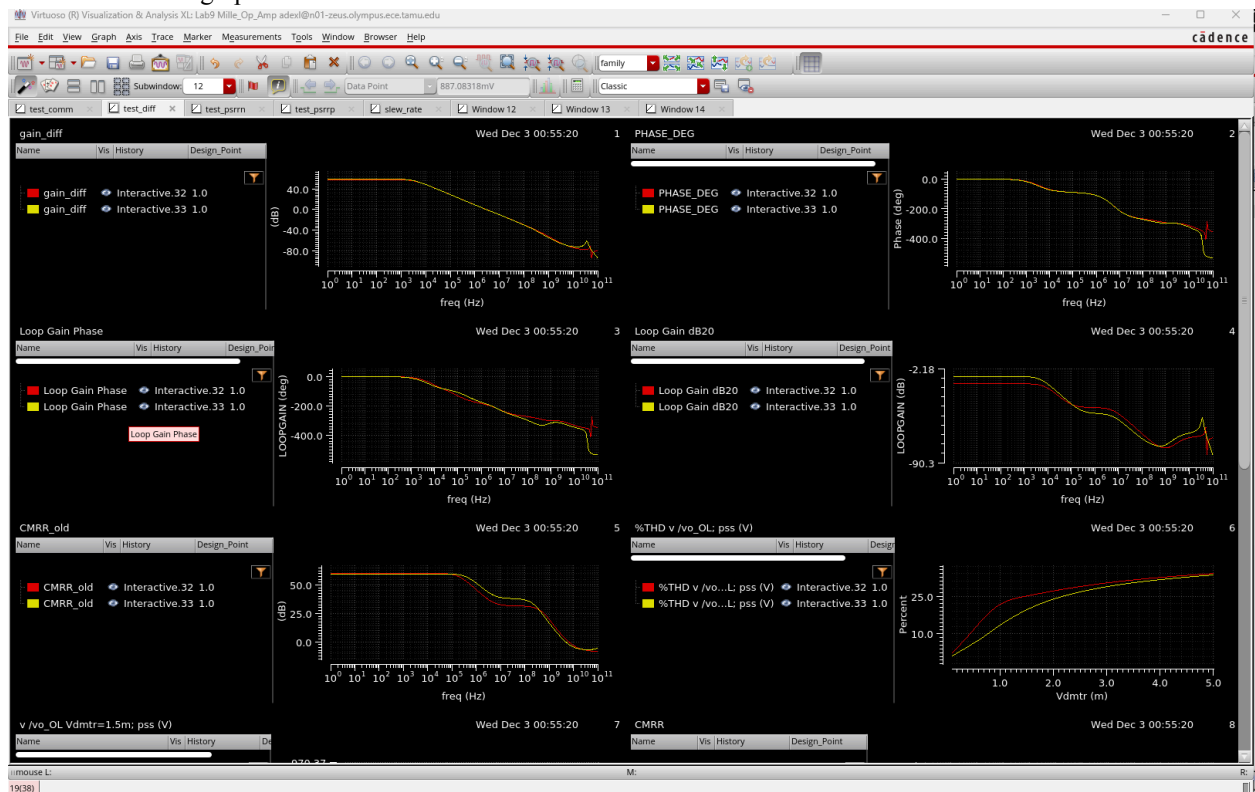
2.COMMON MODE PSS MARGIN:



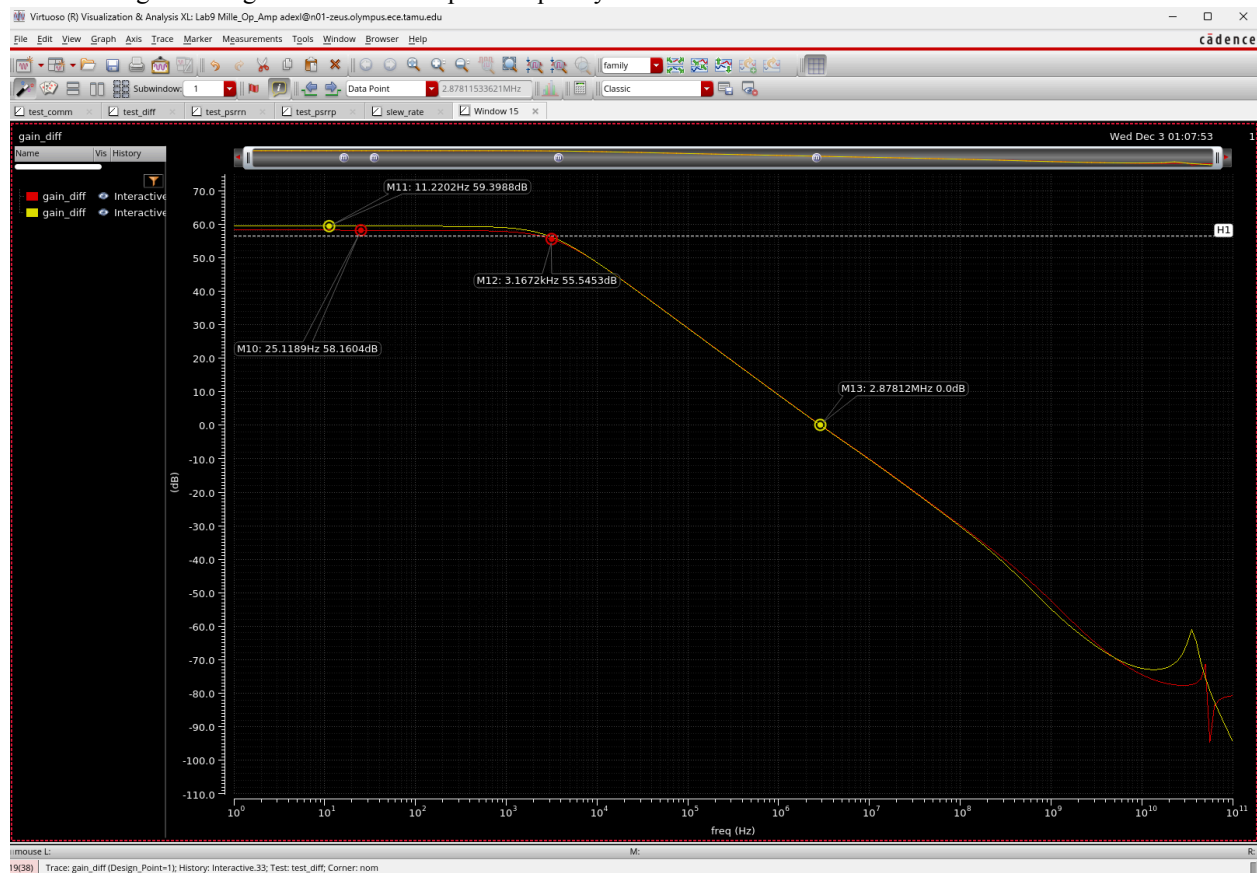
COMMON MODE OUTPUT VOLTAGE SWING:



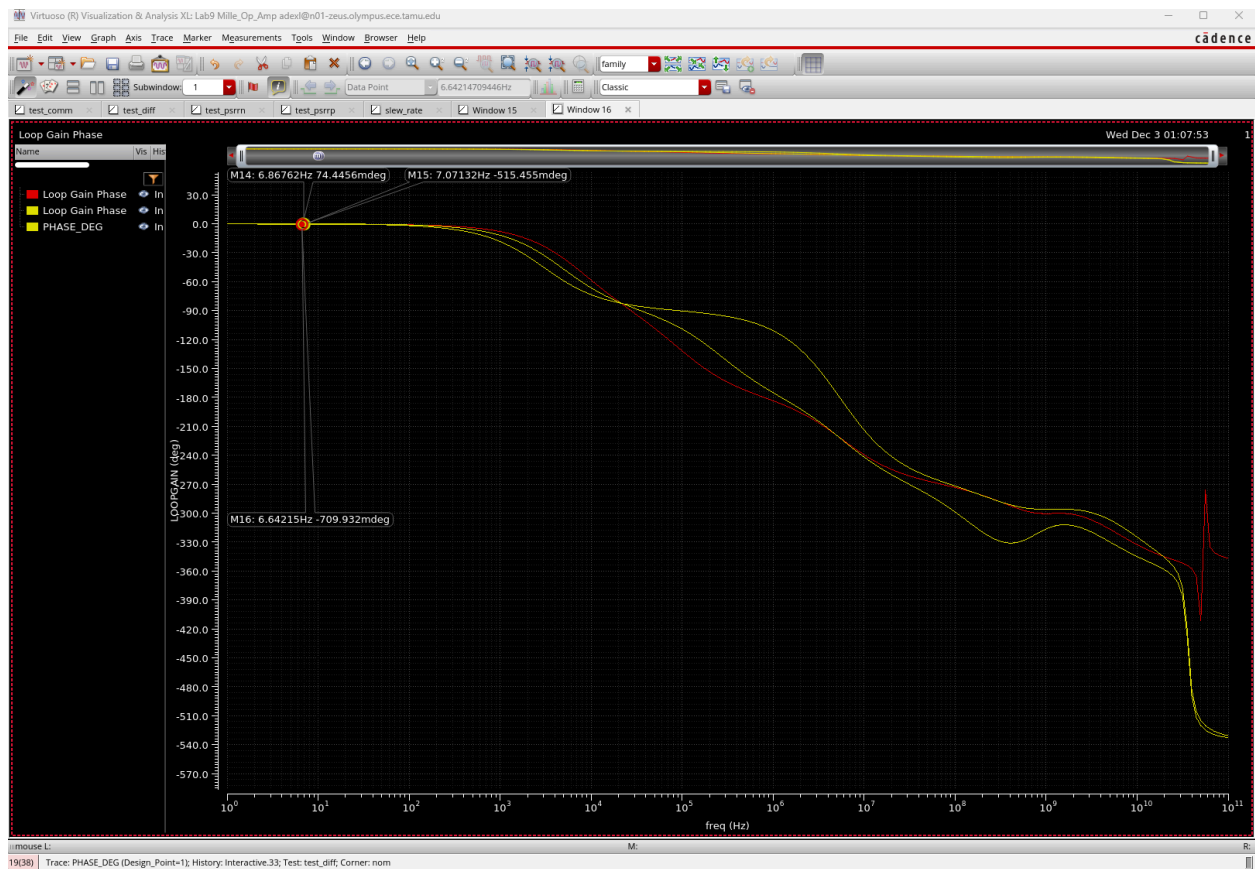
Differential mode graphs:



Differential gain with gbw and dominant pole frequency:



Loop gain with iprobe:

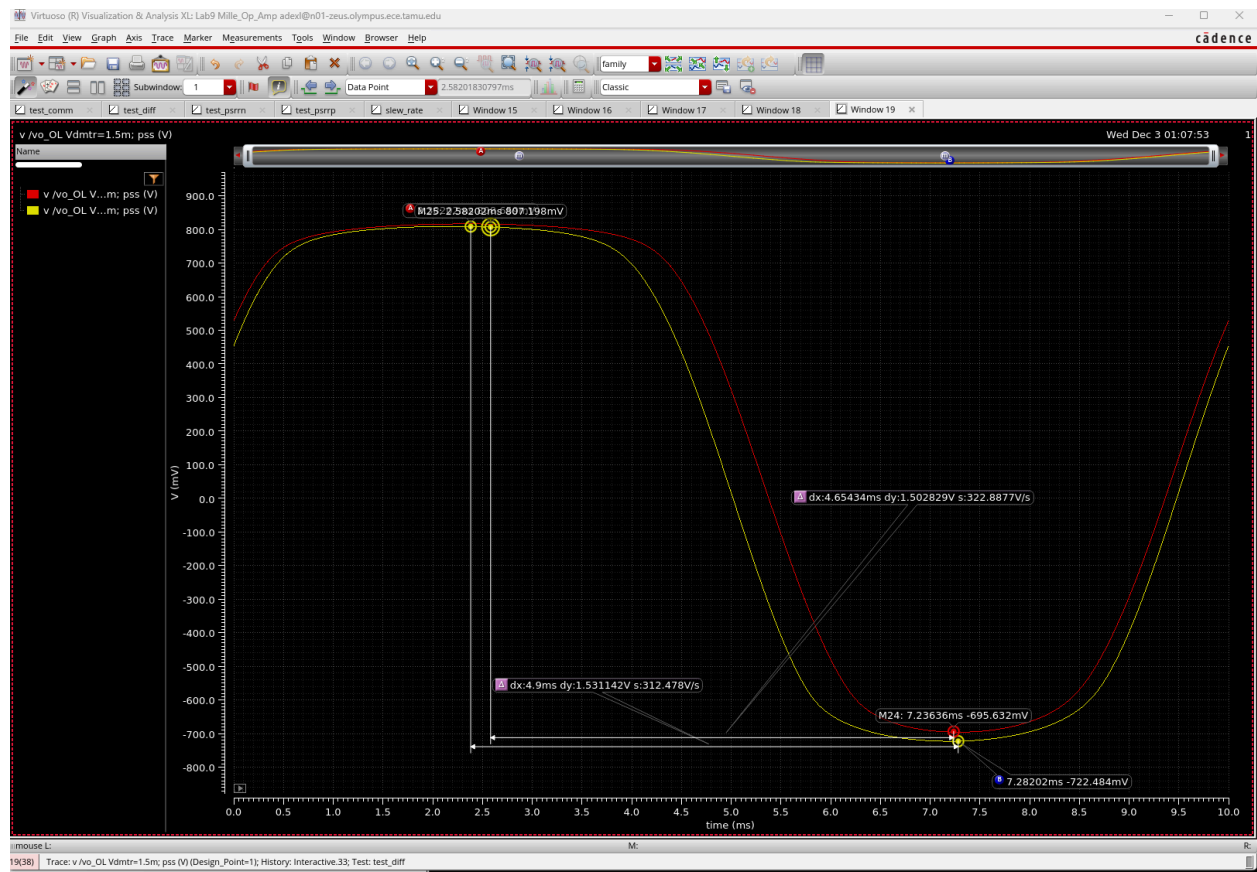


Differential mode CMRR:

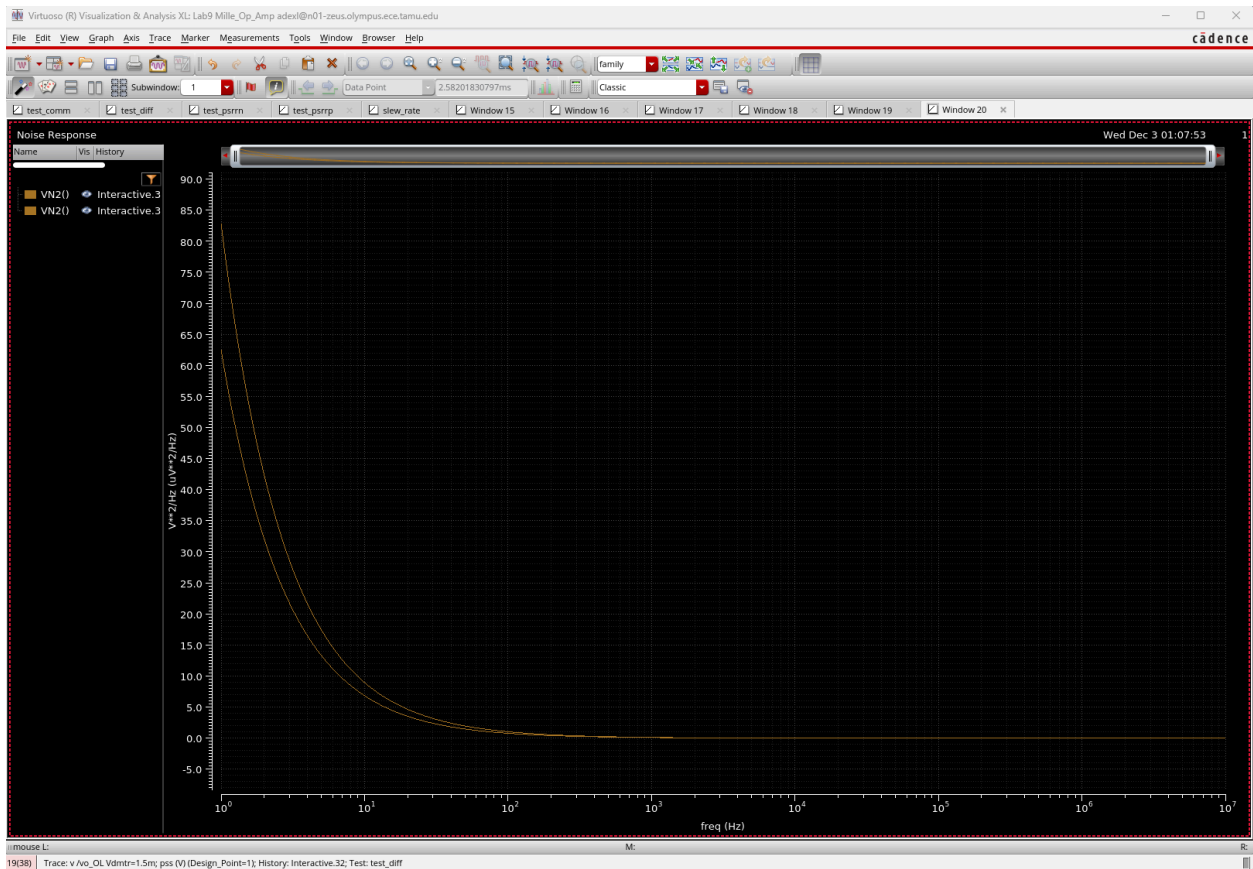




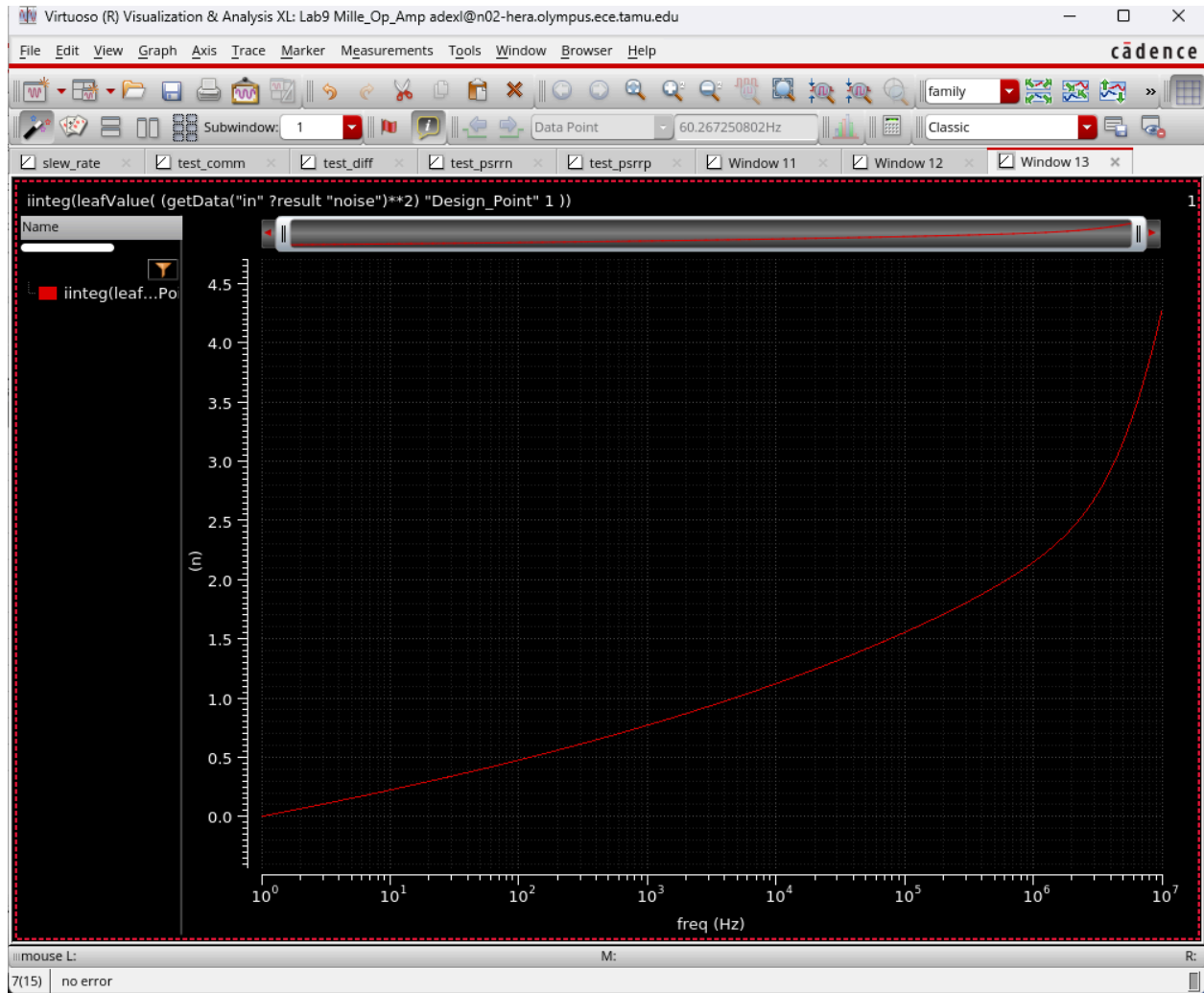
Peak to peak differential voltage swing:

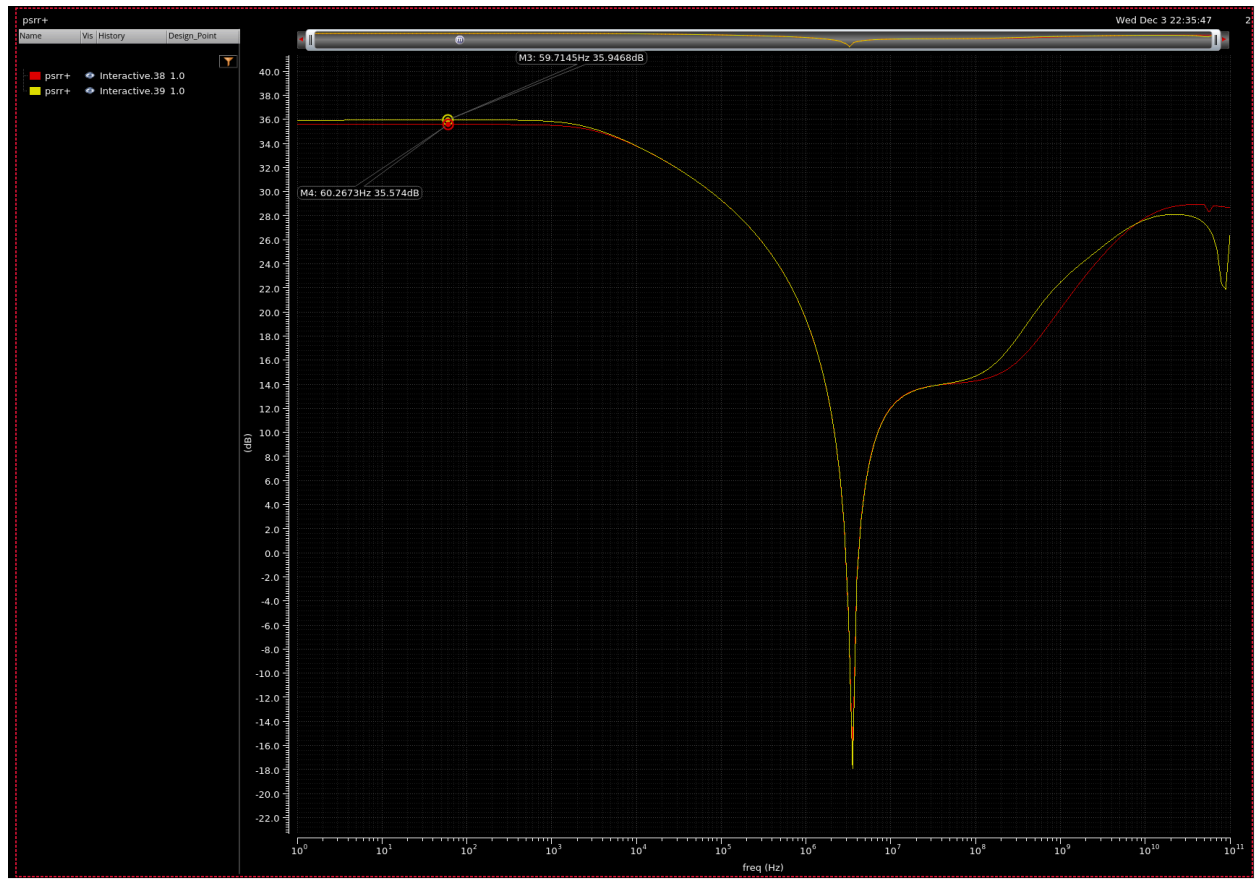


Noise

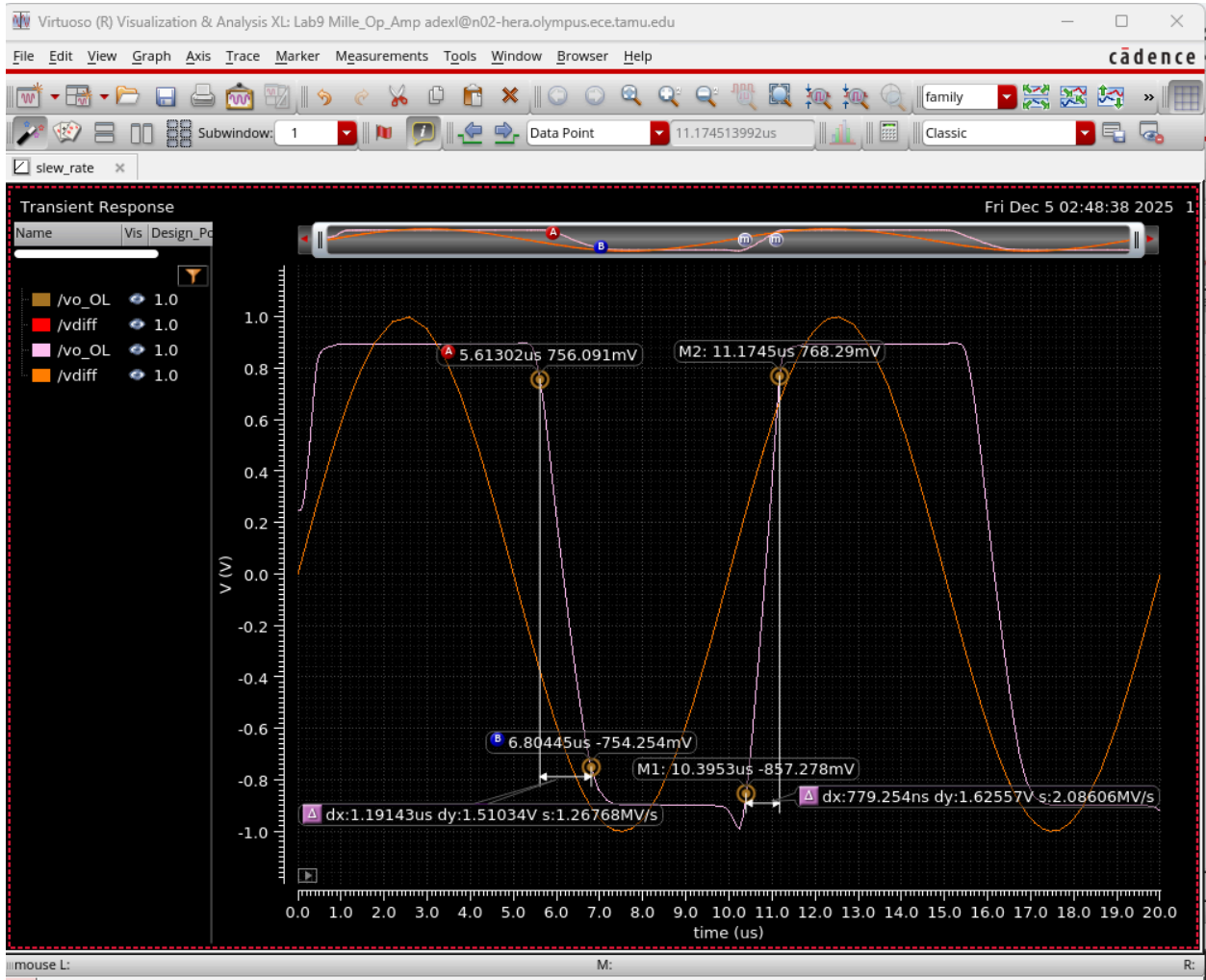


Noise power frequency:





Slew rate for post-layout simulation:



Virtuoso® Analog Design Environment XL Editing: Lab9 Mille_Op_Amp adexl@n02-hera.olympus.ece.tamu.edu

Launch File Create Tools Options Run EAD Parasitics/LDE Window Calibre Help

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No Parasitics/LDE Single Run, Sweeps and Corners Reference:

Data View

- Tests
 - test_comm
 - test_diff
 - Simulator spectre
 - Analyses
 - dc t-0.9 0.9 0.0001 Lin...
 - ac 1 100G 20 Logarith...
 - stb 1 100G 20 /PRB0 L...
 - pss VAR("f_{dm}") 10 0.1...
 - noise 1 10M 100 Logarit...
 - Design Variables
 - test_psrp
 - test_psrn
 - slew_rate
 - Global Variables
 - Parameters
 - Corners
 - Documents

Click to add analysis

Click to add test

Run Summary

1 Test 1 Nominal Corner

1 Point Sweep 4 Corners

History

History Item	Status
Interactive.41	finished with errors
MonteCarlo.2	finished

Outputs Setup Results

Detail

Parameter	Nominal	FastFast	SlowFast	SlowSlow	FastSlow
allModels.scs	tt	ff	sf	ss	fs
design.scs		<unspecified section>	<unspecified section>	<unspecified section>	<unspecified section>

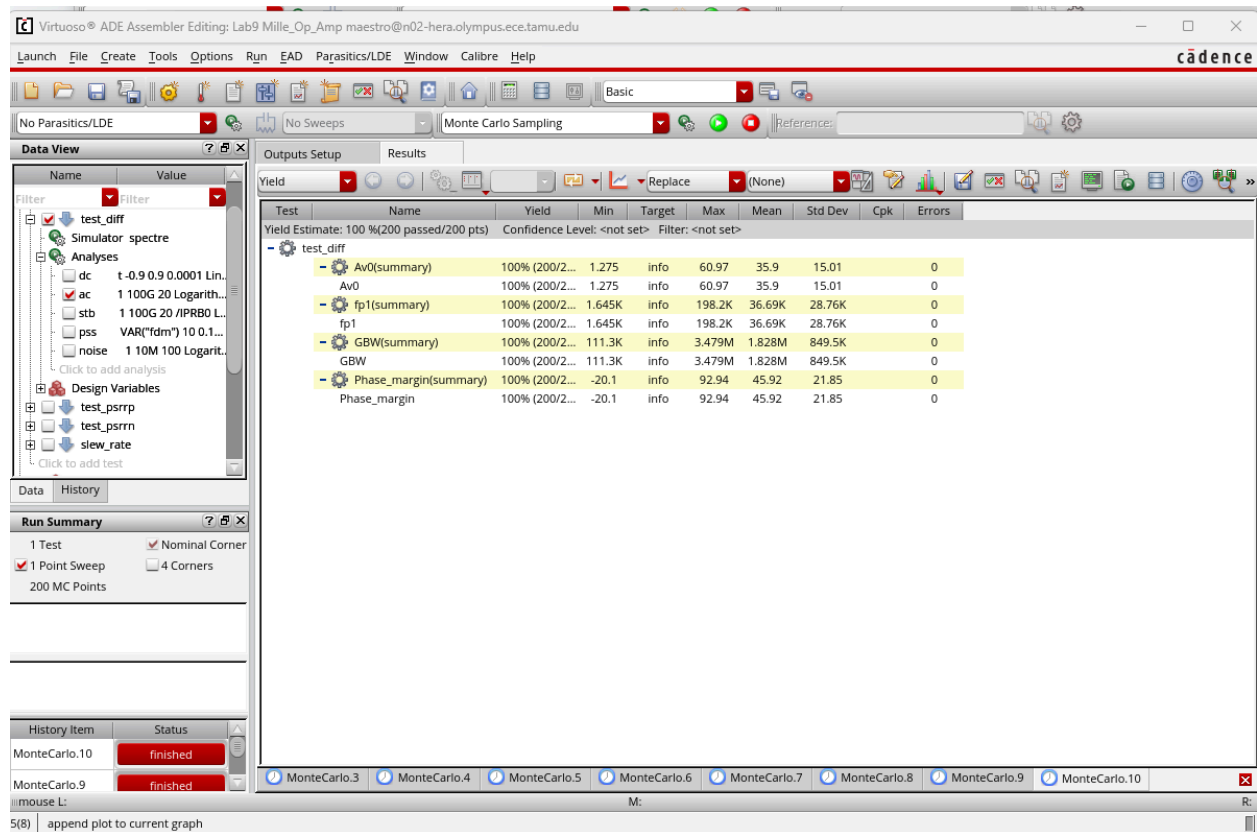
Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	FastFast	SlowFast	SlowSlow	FastSlow
test_diff	/Vo_OL										
test_diff	/CO/PLUS										
test_diff	/IO/VDD										
test_diff	/V3										
test_diff	gain_diff										
test_diff	PHASE_DEG										
test_diff	Loop Gain Phase	eval err						eval err	eval err	eval err	eval err
test_diff	Loop Gain dB20	eval err						eval err	eval err	eval err	eval err
test_diff	CMRR_old										
test_diff	%THD v /Vo_OL; ...	eval err						eval err	eval err	eval err	eval err
test_diff	v /Vo_OL Vdmtr...	eval err						eval err	eval err	eval err	eval err
test_diff	v /Vo_OL Vdmtr...	eval err						eval err	eval err	eval err	eval err
test_diff	CMRR	eval err						eval err	eval err	eval err	eval err
test_diff	input noise; V**...	eval err						eval err	eval err	eval err	eval err
test_diff	/vdiff										
test_diff	Av0	59.5				58.39	60.74	58.39	59.93	60.74	59.08
test_diff	fp1	2.917K				1.831K	4.69K	4.69K	2.347K	1.831K	3.475K
test_diff	/net8										
test_diff	GBW	2.878M				2.153M	3.954M	3.954M	2.437M	2.153M	3.279M
test_diff	Phase_margin	33.93				31.72	36.5	31.72	36.5	34.81	32.05
test_diff	/net6										

Interactive.41

mouse L: M: R:

7(11) Build Parasitic/LDE View

MONTE CARLO SIMULATION:



Histogram for Monte carlo simulation:

